

BFS Traversal

Step 1: Start at E

Legend

Complete

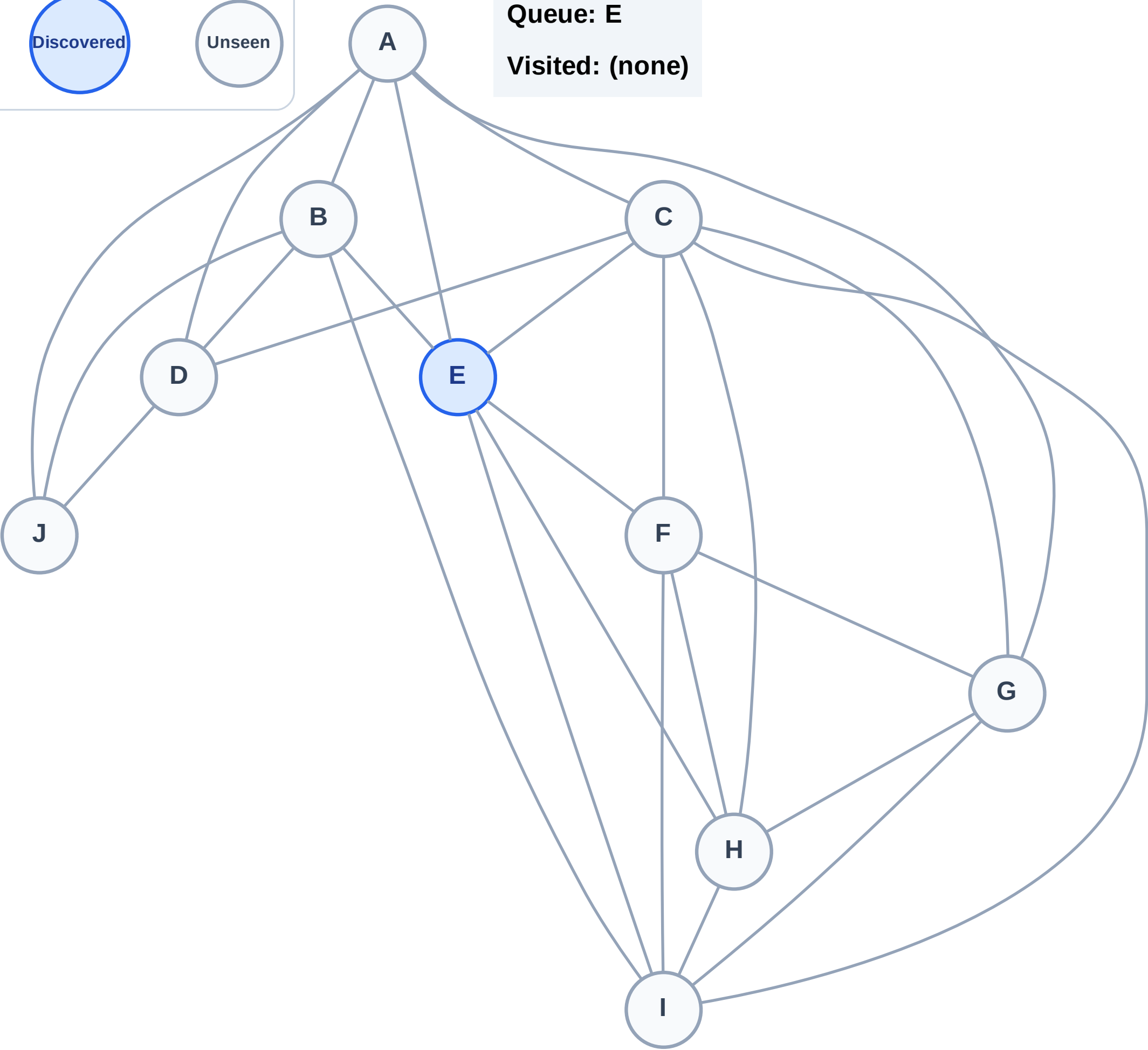
Processing

Discovered

Unseen

Queue: E

Visited: (none)



BFS Traversal

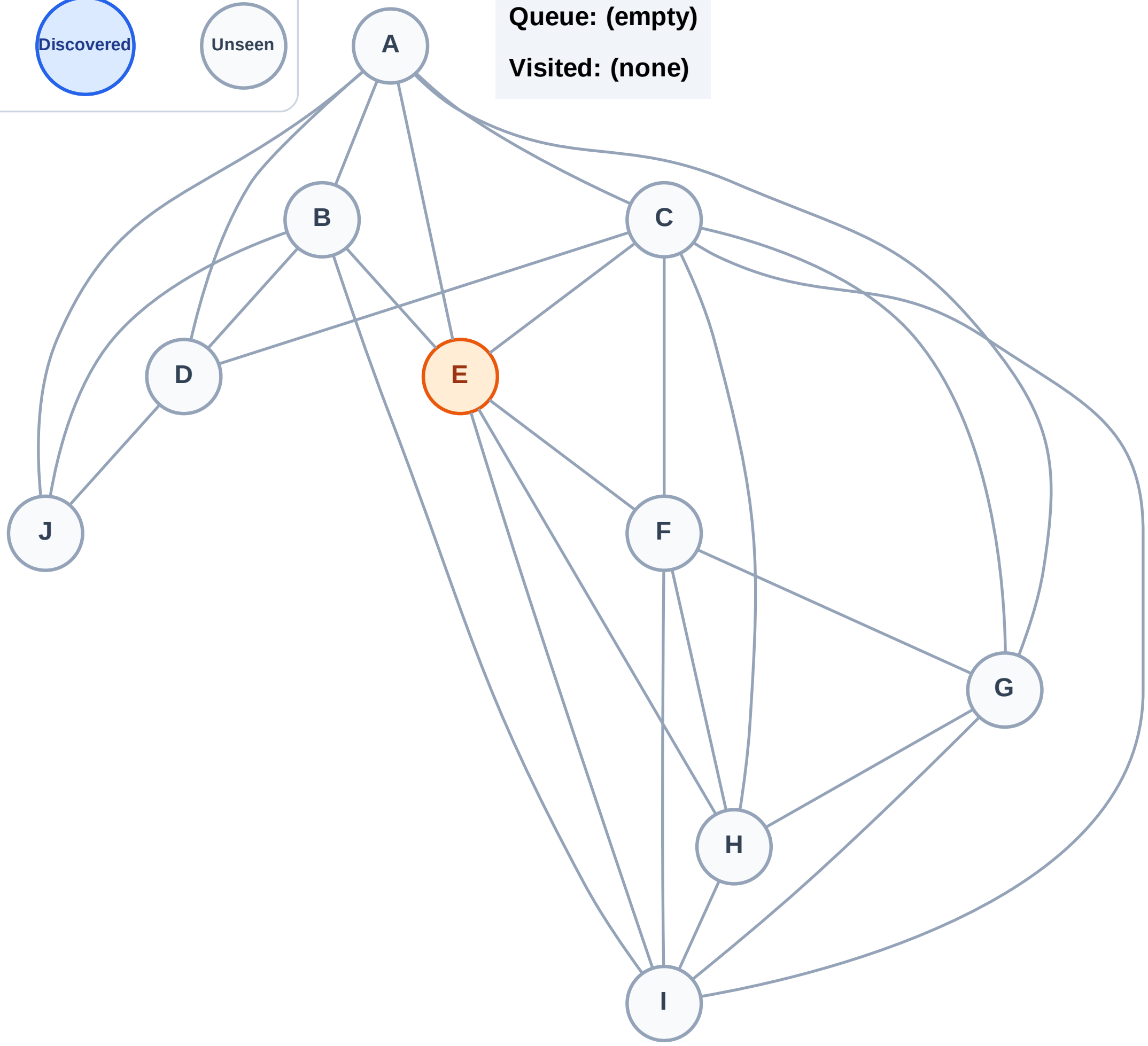
Step 2: Process node E

Legend



Queue: (empty)

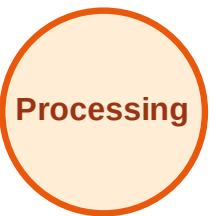
Visited: (none)



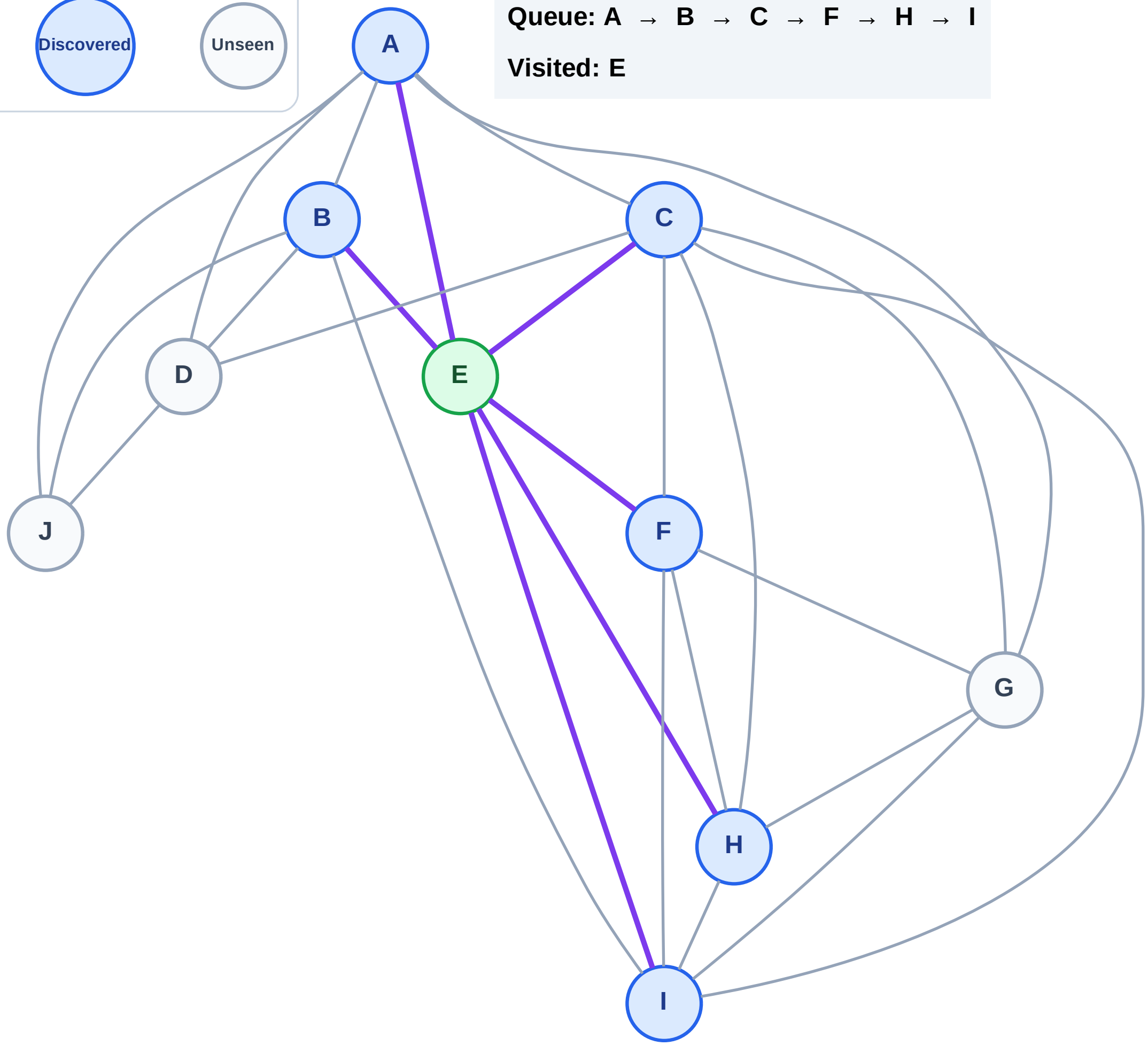
BFS Traversal

Step 3: Discover A, B, C, F, H, I from E

Legend



Queue: A → B → C → F → H → I
Visited: E



BFS Traversal

Step 4: Process node A

Legend

Complete

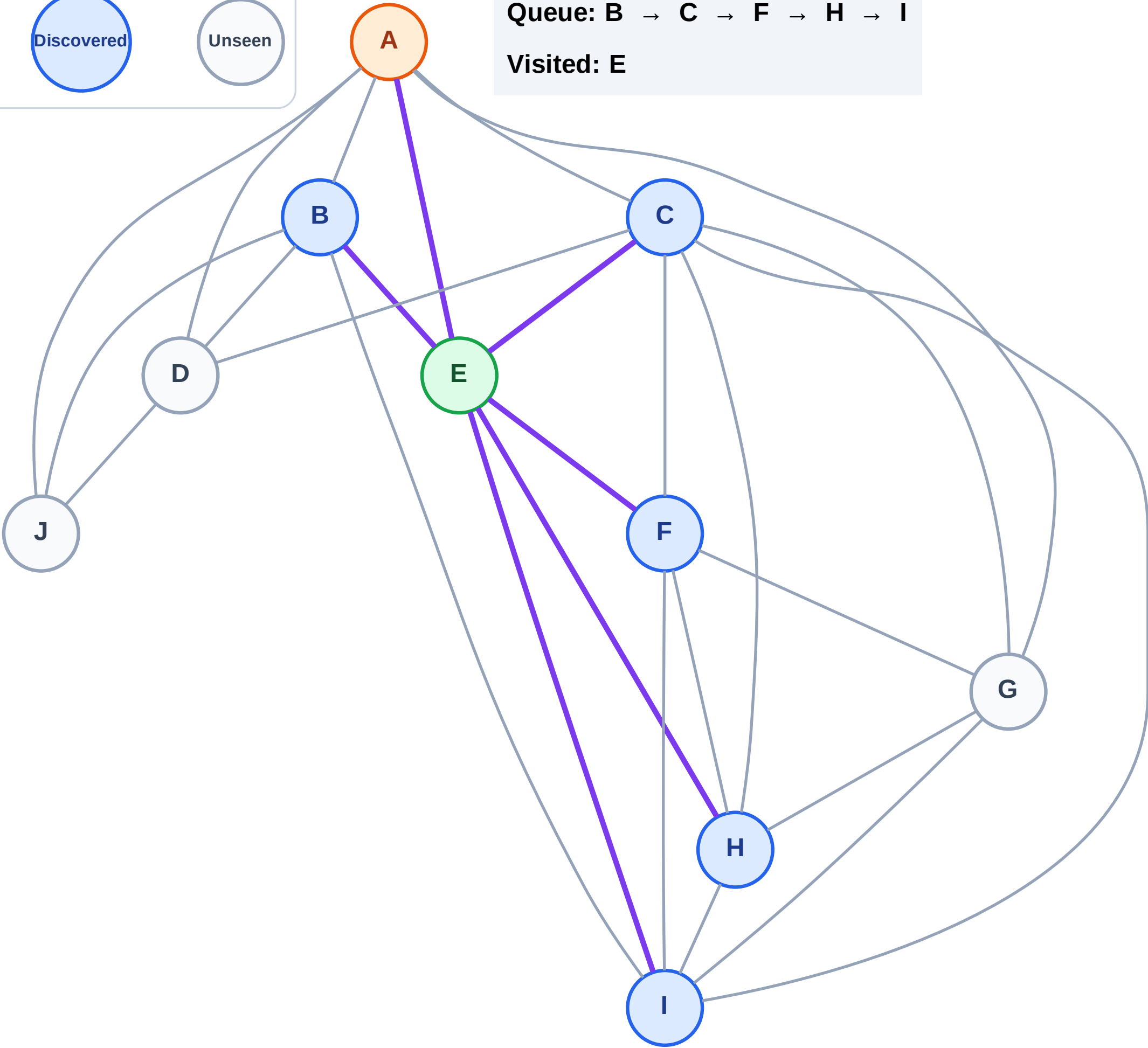
Processing

Discovered

Unseen

Queue: B → C → F → H → I

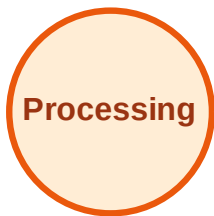
Visited: E



BFS Traversal

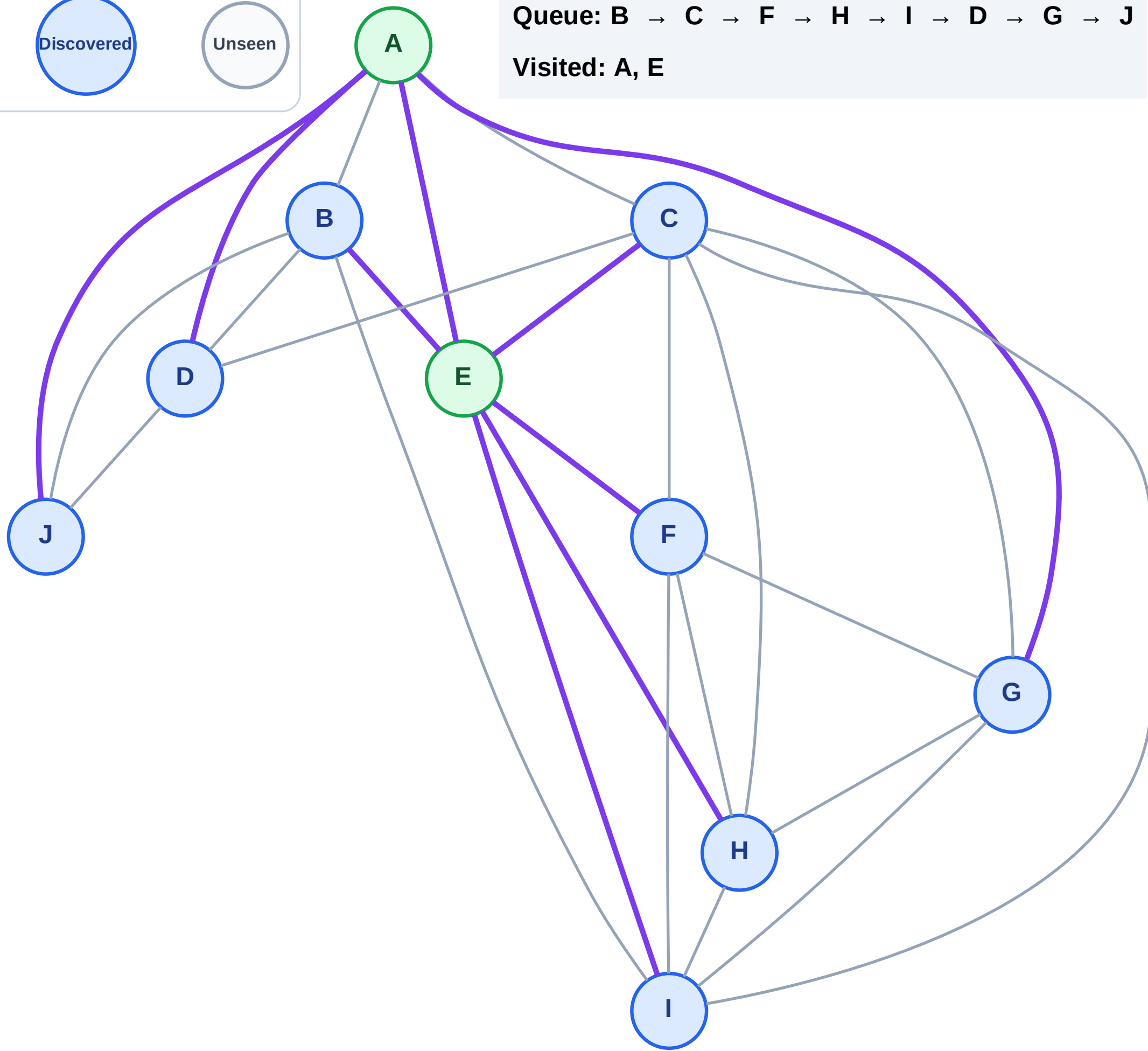
Step 5: Discover D, G, J from A

Legend



Queue: B → C → F → H → I → D → G → J

Visited: A, E



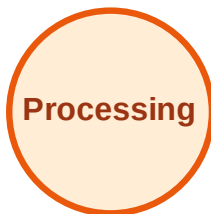
BFS Traversal

Step 6: Process node B

Legend



Complete



Processing



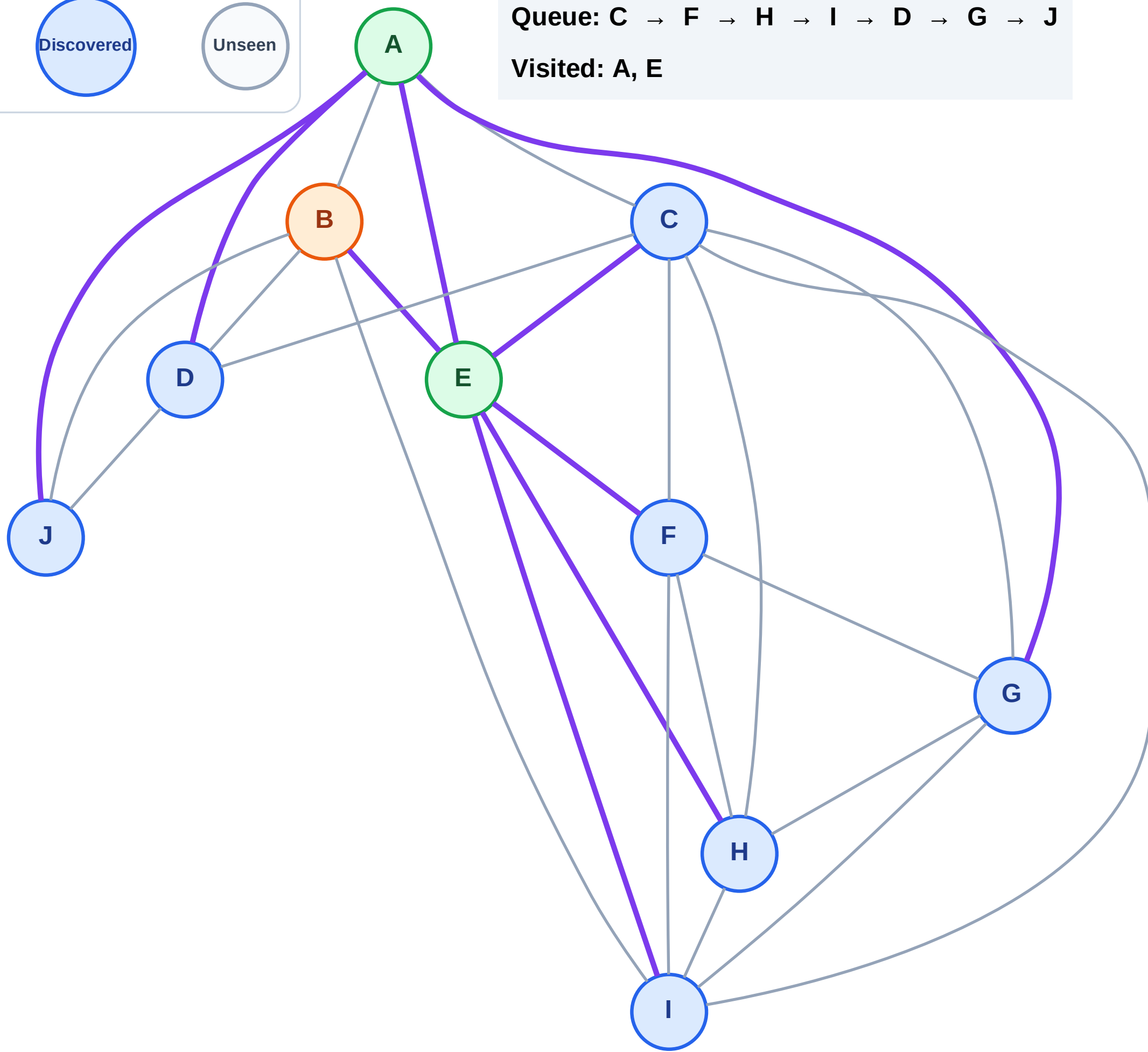
Discovered



Unseen

Queue: C → F → H → I → D → G → J

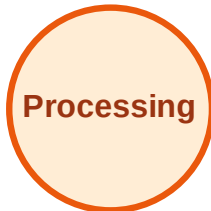
Visited: A, E



BFS Traversal

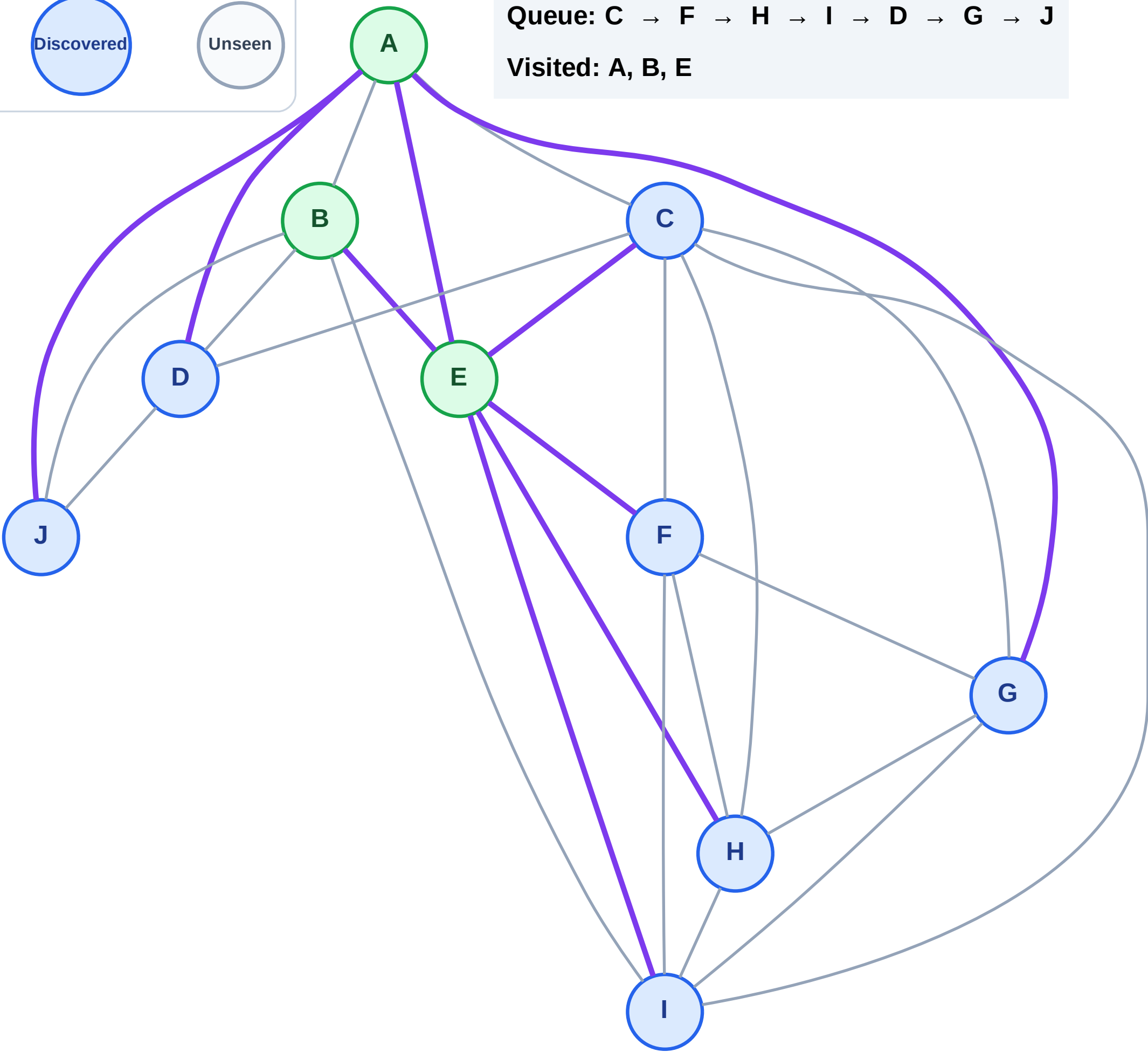
Step 7: No new neighbors from B

Legend



Queue: C → F → H → I → D → G → J

Visited: A, B, E



BFS Traversal

Step 8: Process node C

Legend

Complete

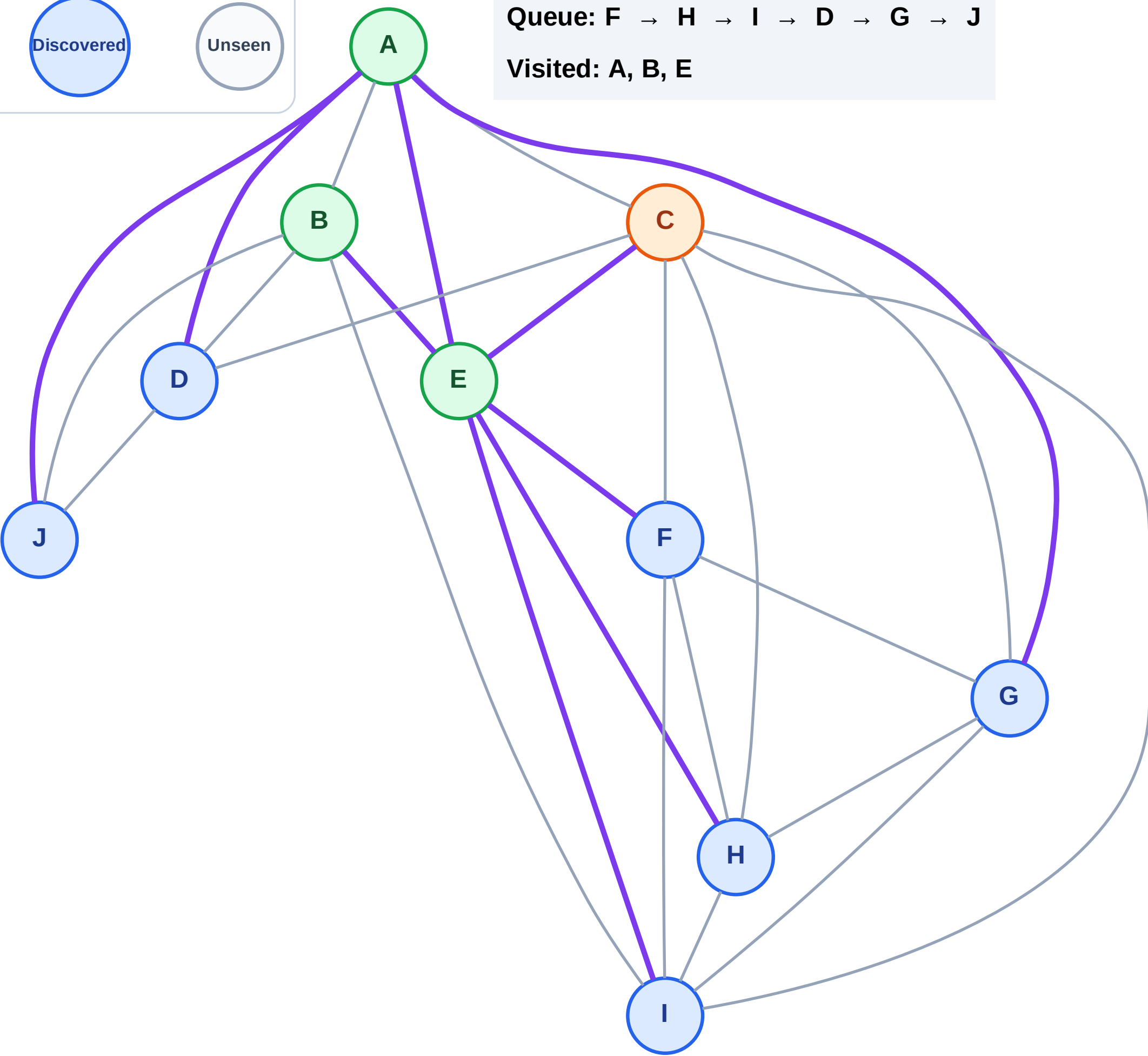
Processing

Discovered

Unseen

Queue: F → H → I → D → G → J

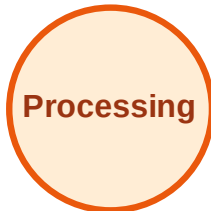
Visited: A, B, E



BFS Traversal

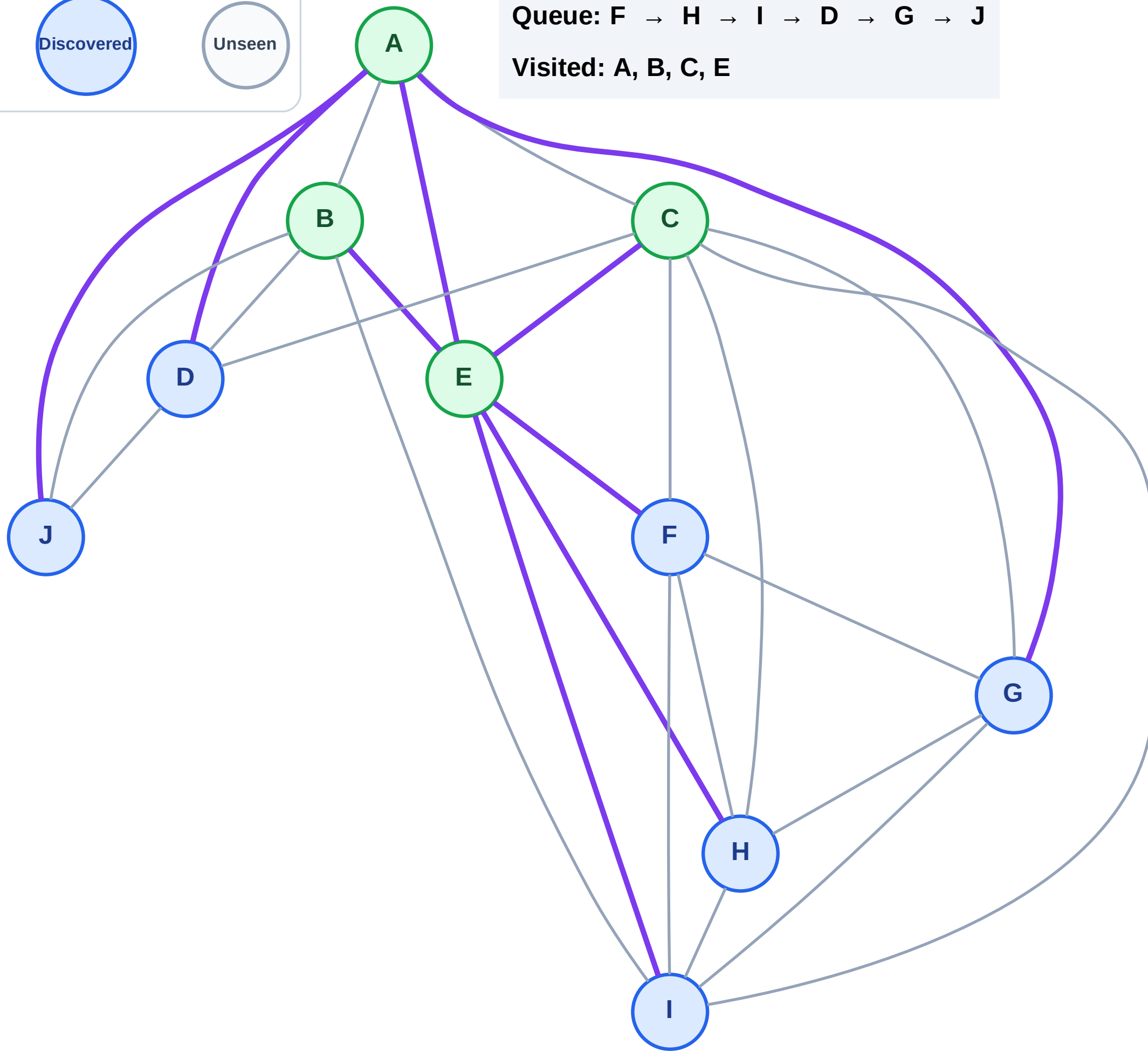
Step 9: No new neighbors from C

Legend



Queue: F → H → I → D → G → J

Visited: A, B, C, E



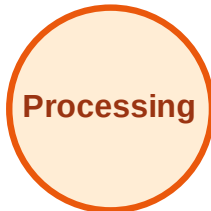
BFS Traversal

Step 10: Process node F

Legend



Complete



Processing



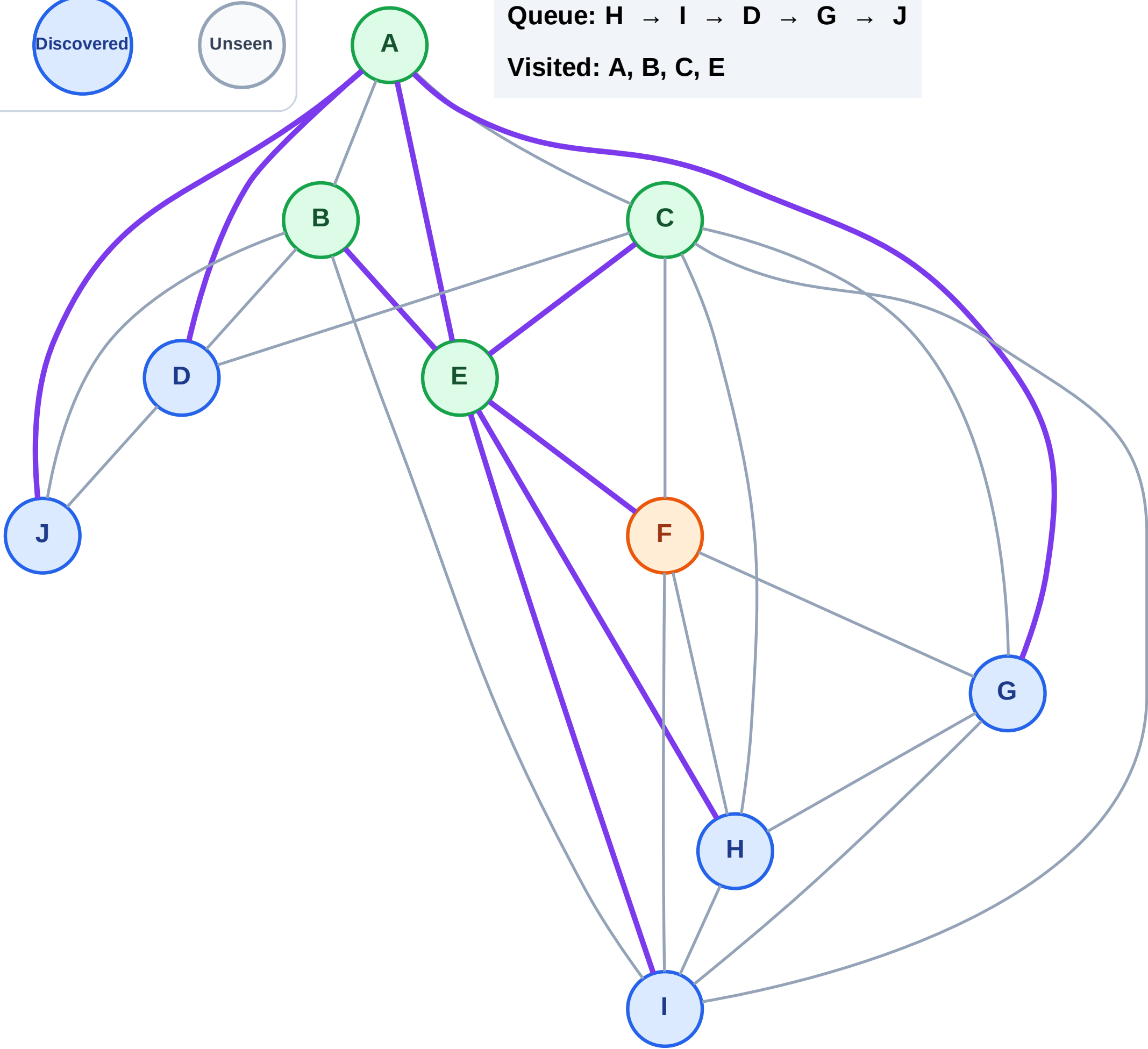
Discovered



Unseen

Queue: H → I → D → G → J

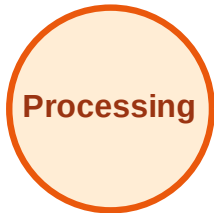
Visited: A, B, C, E



BFS Traversal

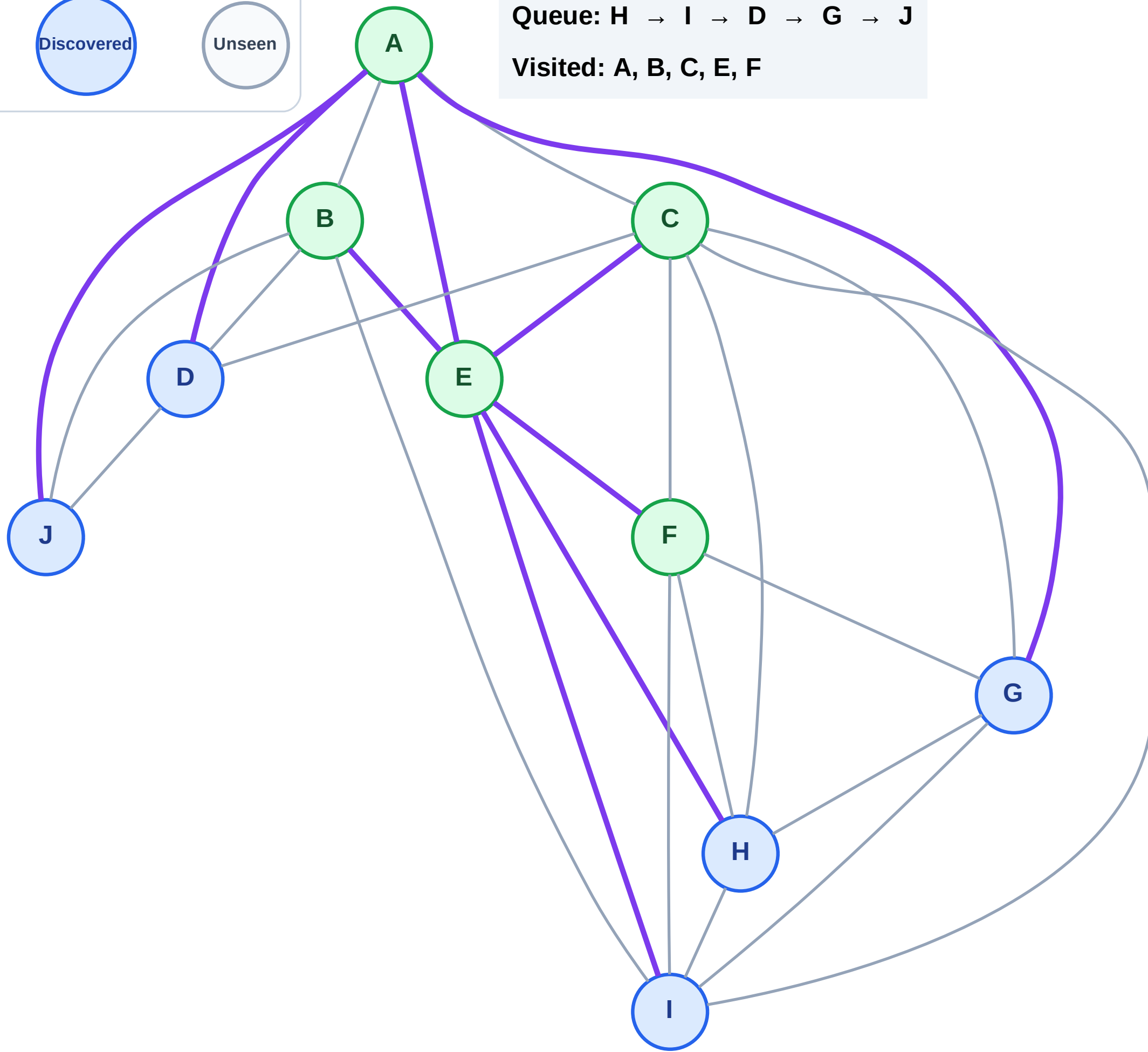
Step 11: No new neighbors from F

Legend



Queue: H → I → D → G → J

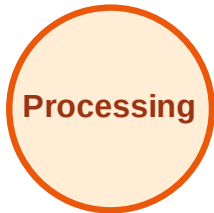
Visited: A, B, C, E, F



BFS Traversal

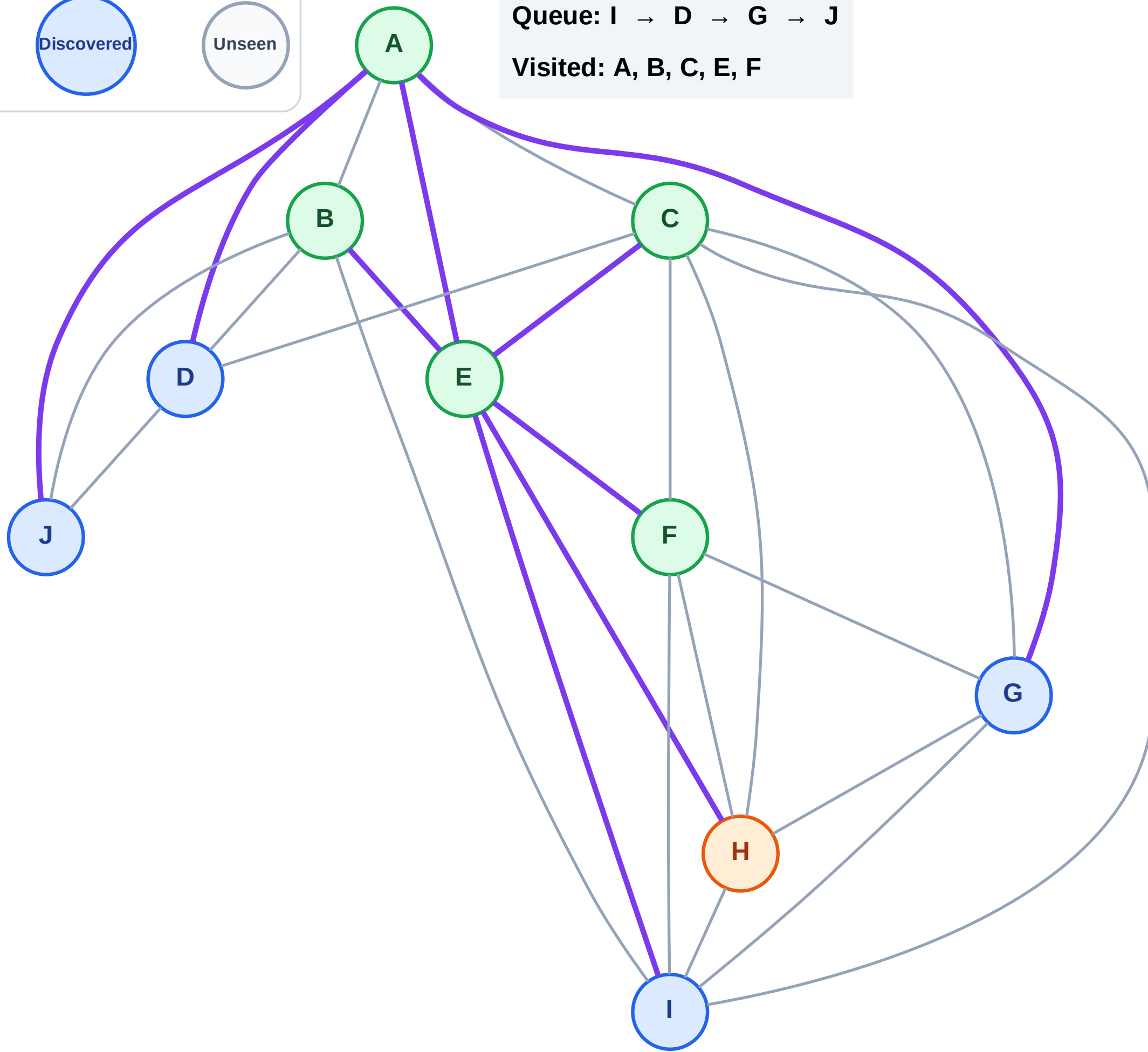
Step 12: Process node H

Legend



Queue: I → D → G → J

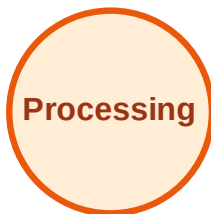
Visited: A, B, C, E, F



BFS Traversal

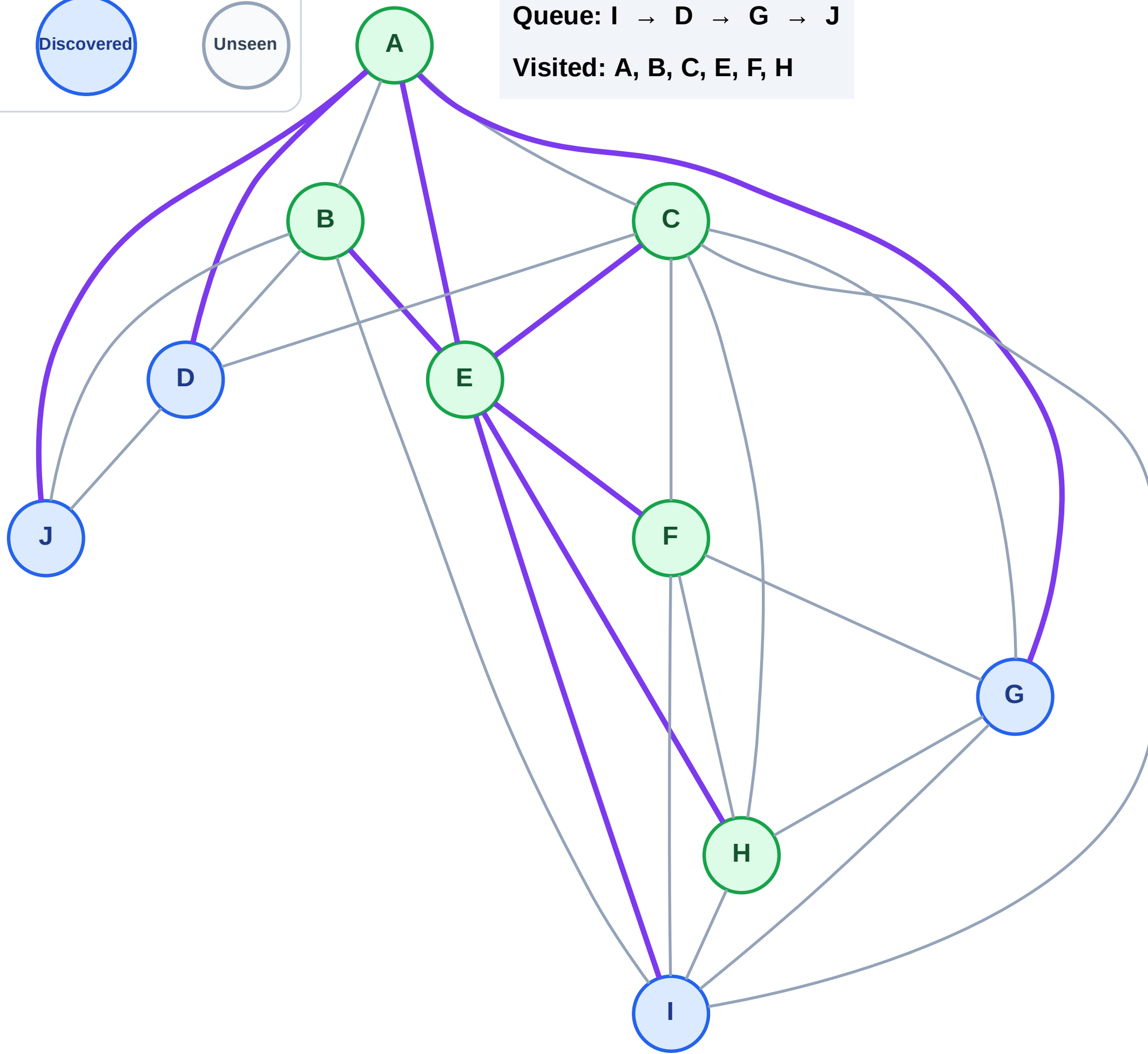
Step 13: No new neighbors from H

Legend



Queue: I → D → G → J

Visited: A, B, C, E, F, H



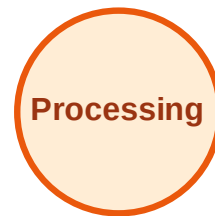
BFS Traversal

Step 14: Process node I

Legend



Complete



Processing

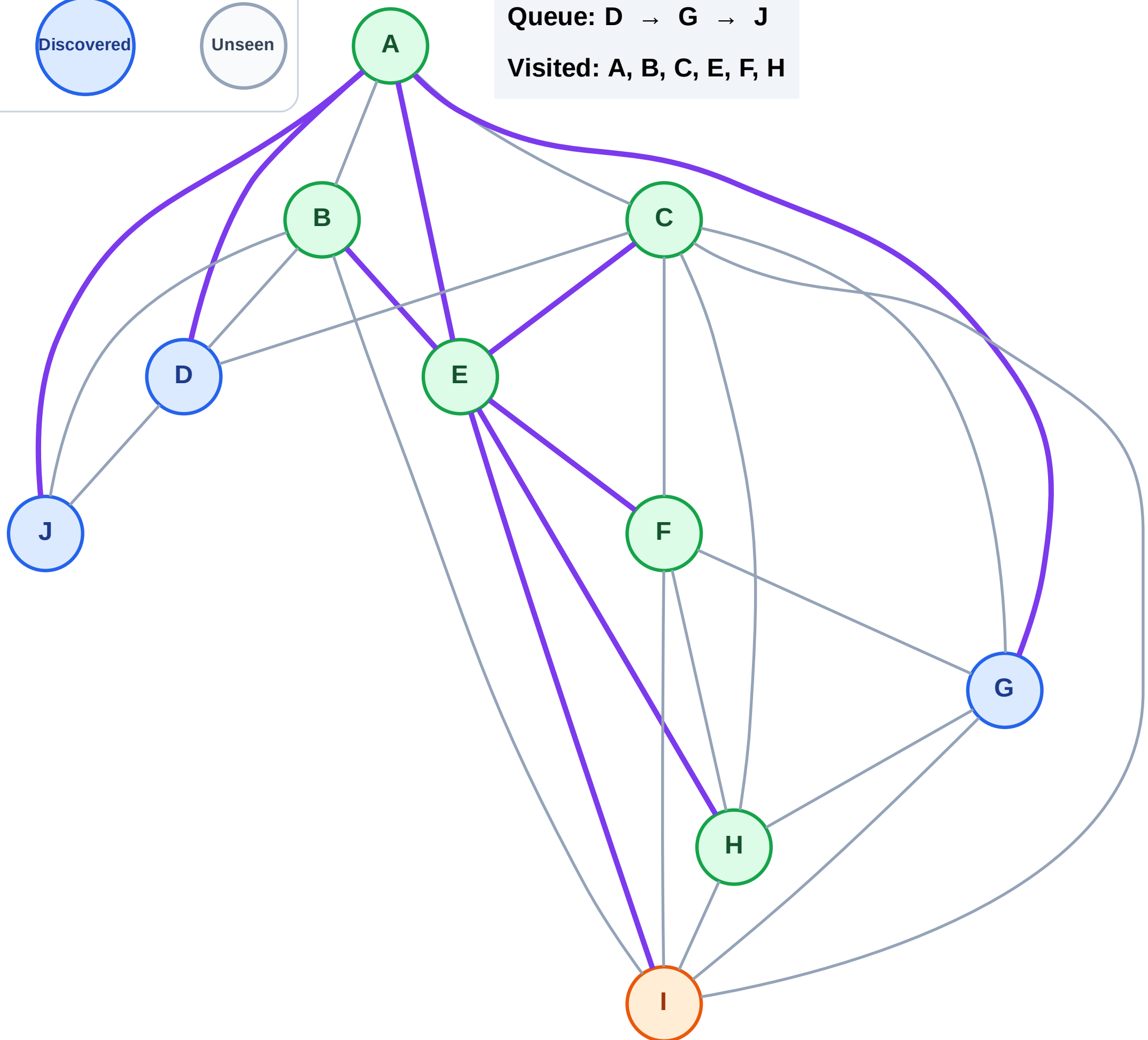


Discovered



Unseen

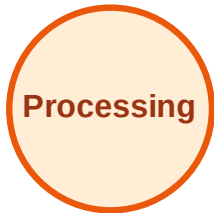
Queue: D → G → J
Visited: A, B, C, E, F, H



BFS Traversal

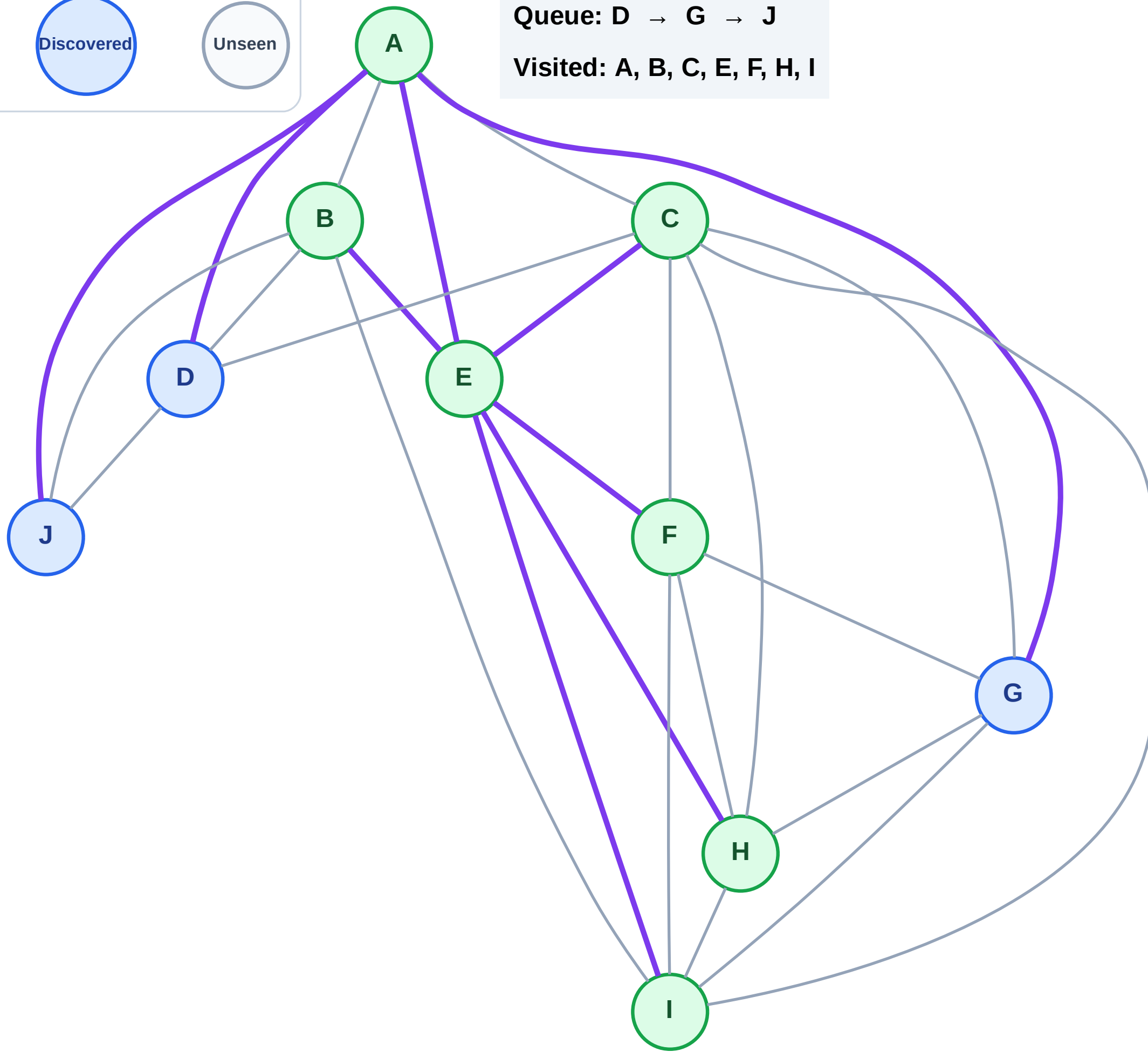
Step 15: No new neighbors from I

Legend



Queue: D → G → J

Visited: A, B, C, E, F, H, I



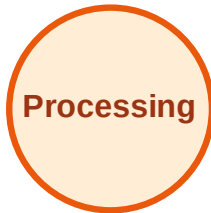
BFS Traversal

Step 16: Process node D

Legend



Complete



Processing



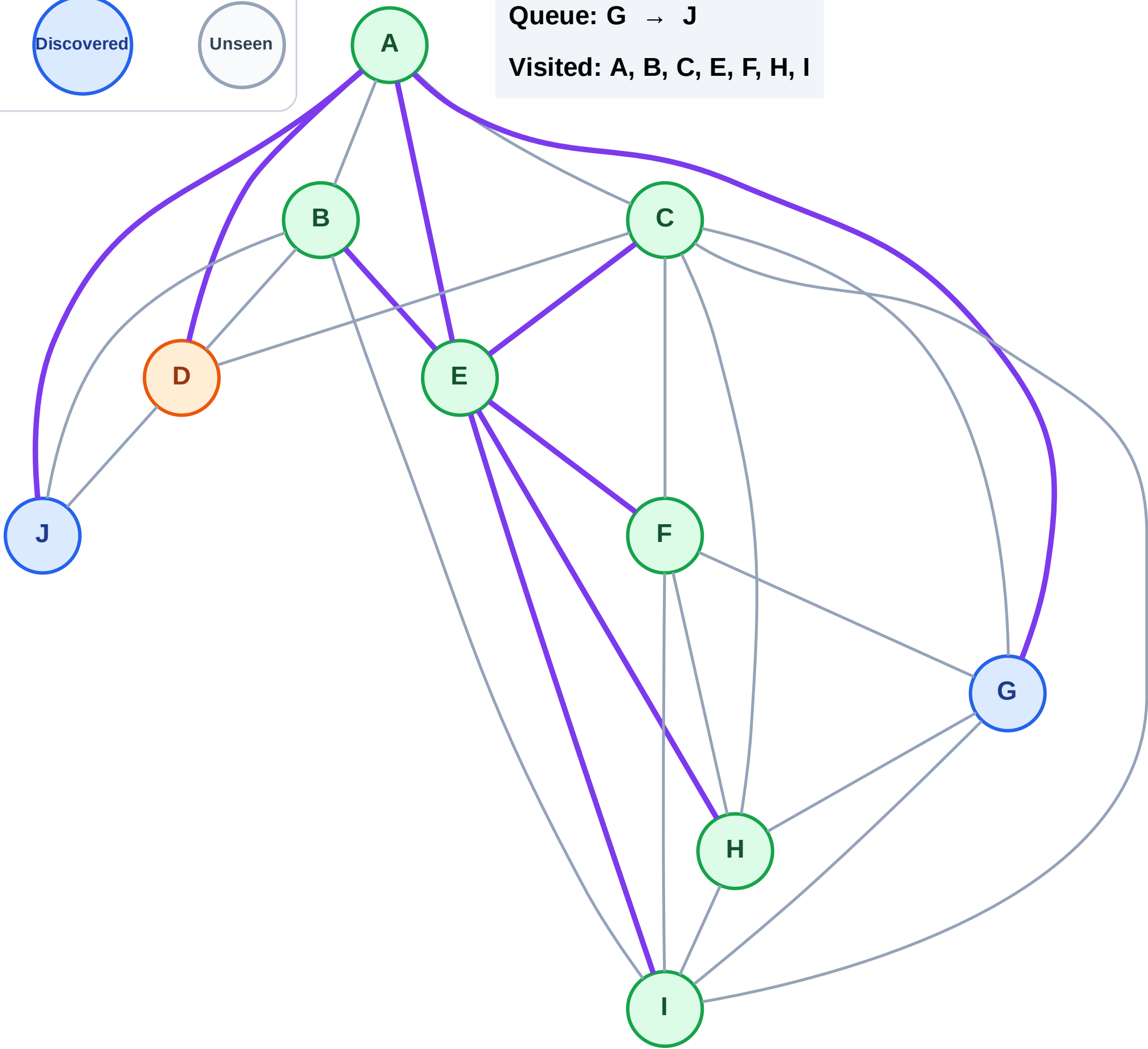
Discovered



Unseen

Queue: G → J

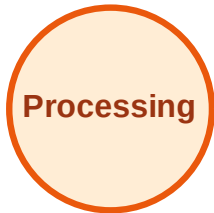
Visited: A, B, C, E, F, H, I



BFS Traversal

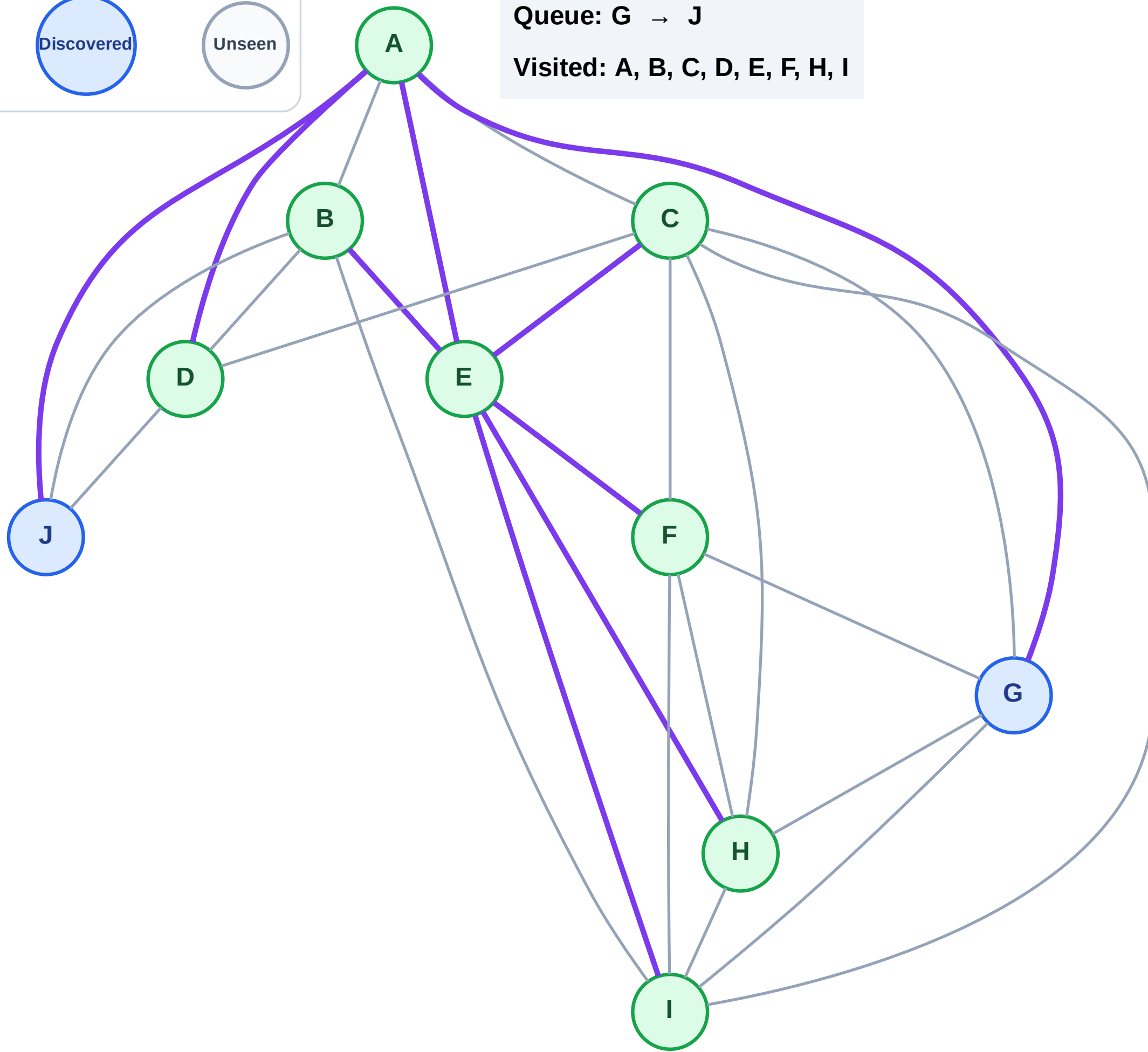
Step 17: No new neighbors from D

Legend



Queue: G → J

Visited: A, B, C, D, E, F, H, I



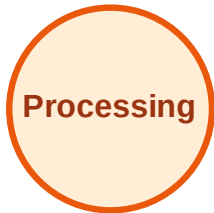
BFS Traversal

Step 18: Process node G

Legend



Complete



Processing



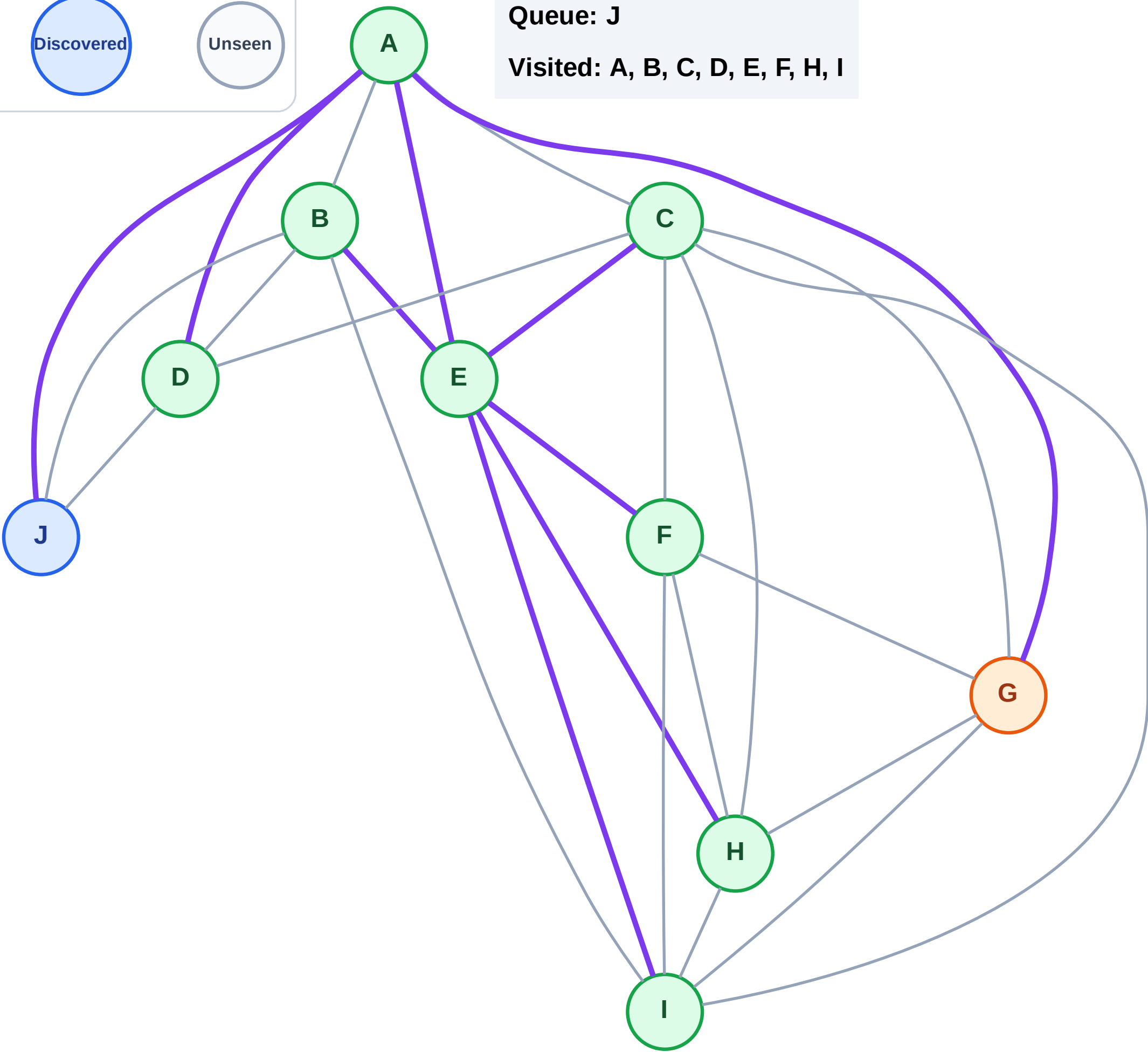
Discovered



Unseen

Queue: J

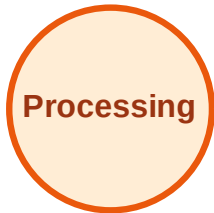
Visited: A, B, C, D, E, F, H, I



BFS Traversal

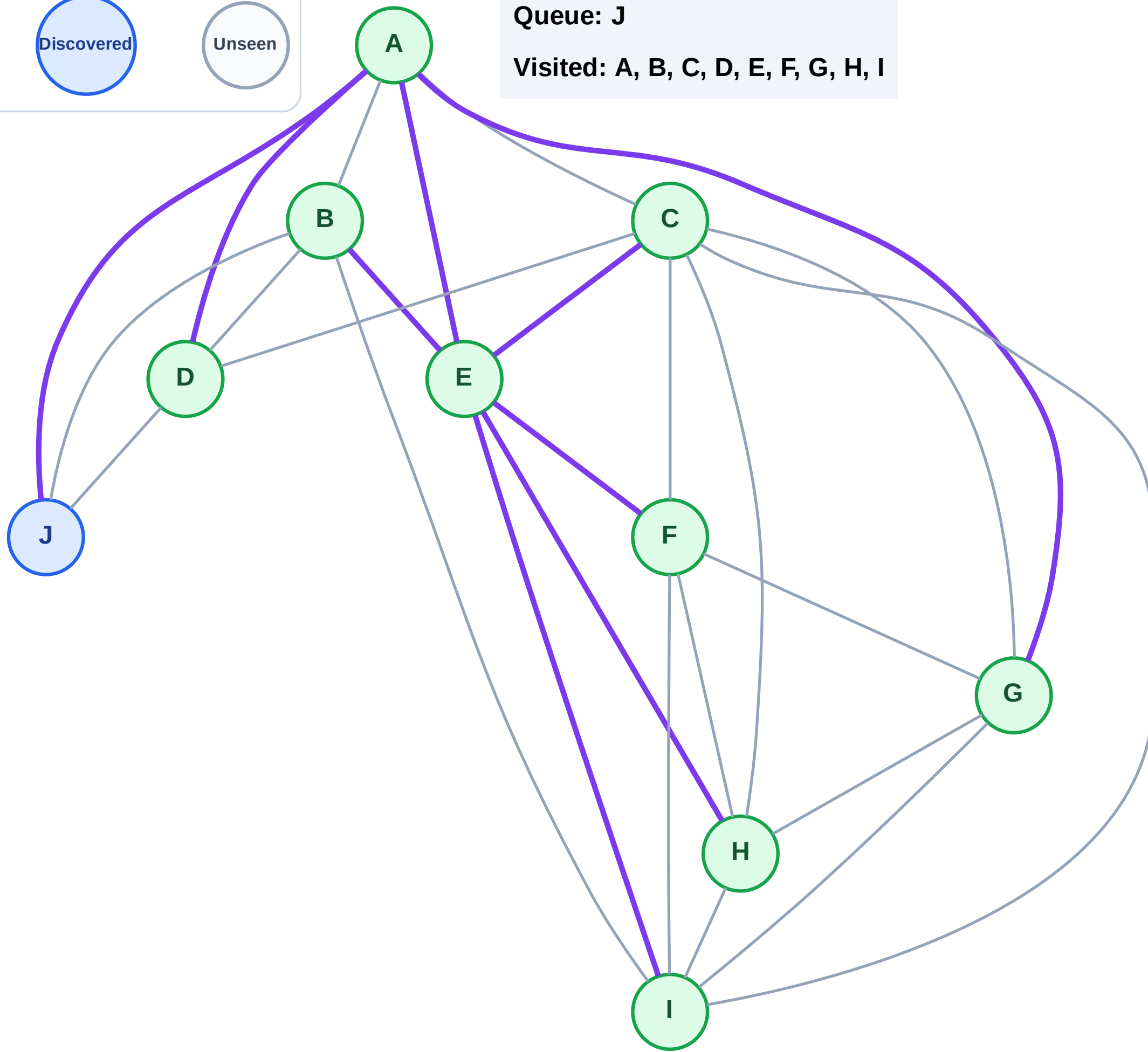
Step 19: No new neighbors from G

Legend



Queue: J

Visited: A, B, C, D, E, F, G, H, I



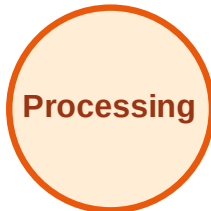
BFS Traversal

Step 20: Process node J

Legend



Complete



Processing



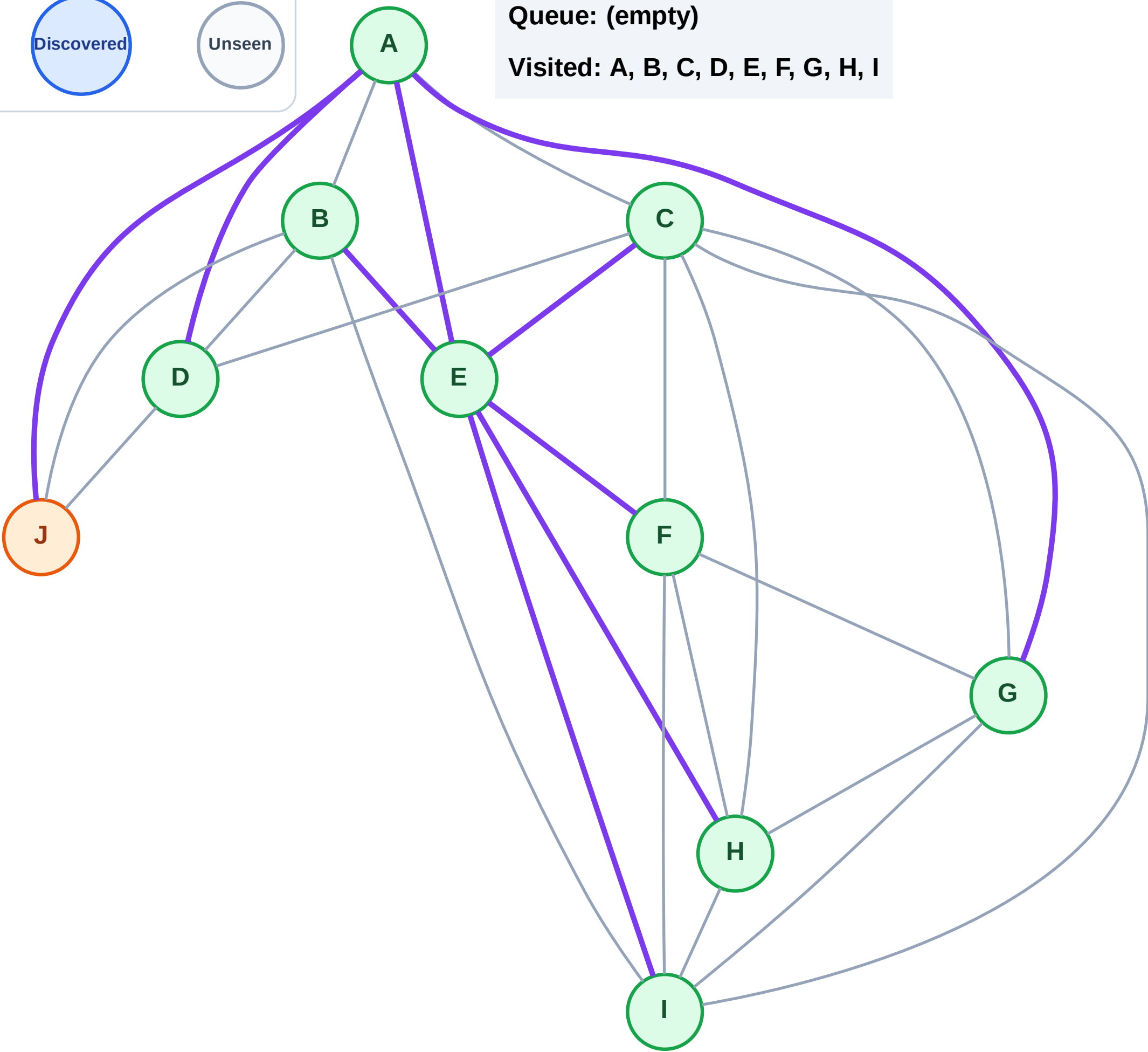
Discovered



Unseen

Queue: (empty)

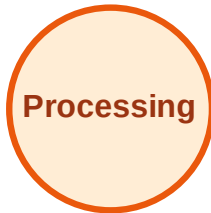
Visited: A, B, C, D, E, F, G, H, I



BFS Traversal

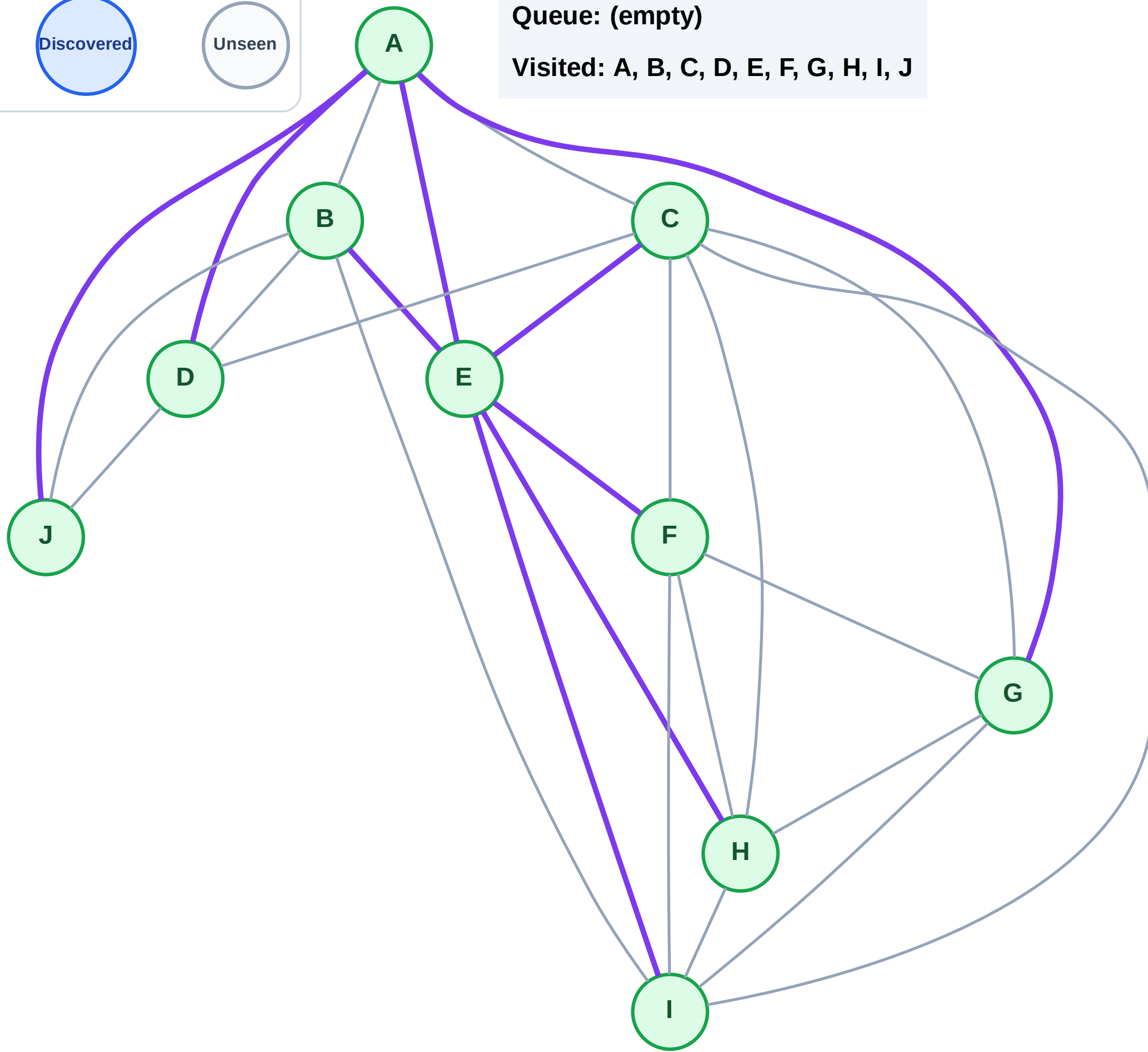
Step 21: No new neighbors from J

Legend



Queue: (empty)

Visited: A, B, C, D, E, F, G, H, I, J



DFS Traversal

Step 1: Start at E

Legend

Complete

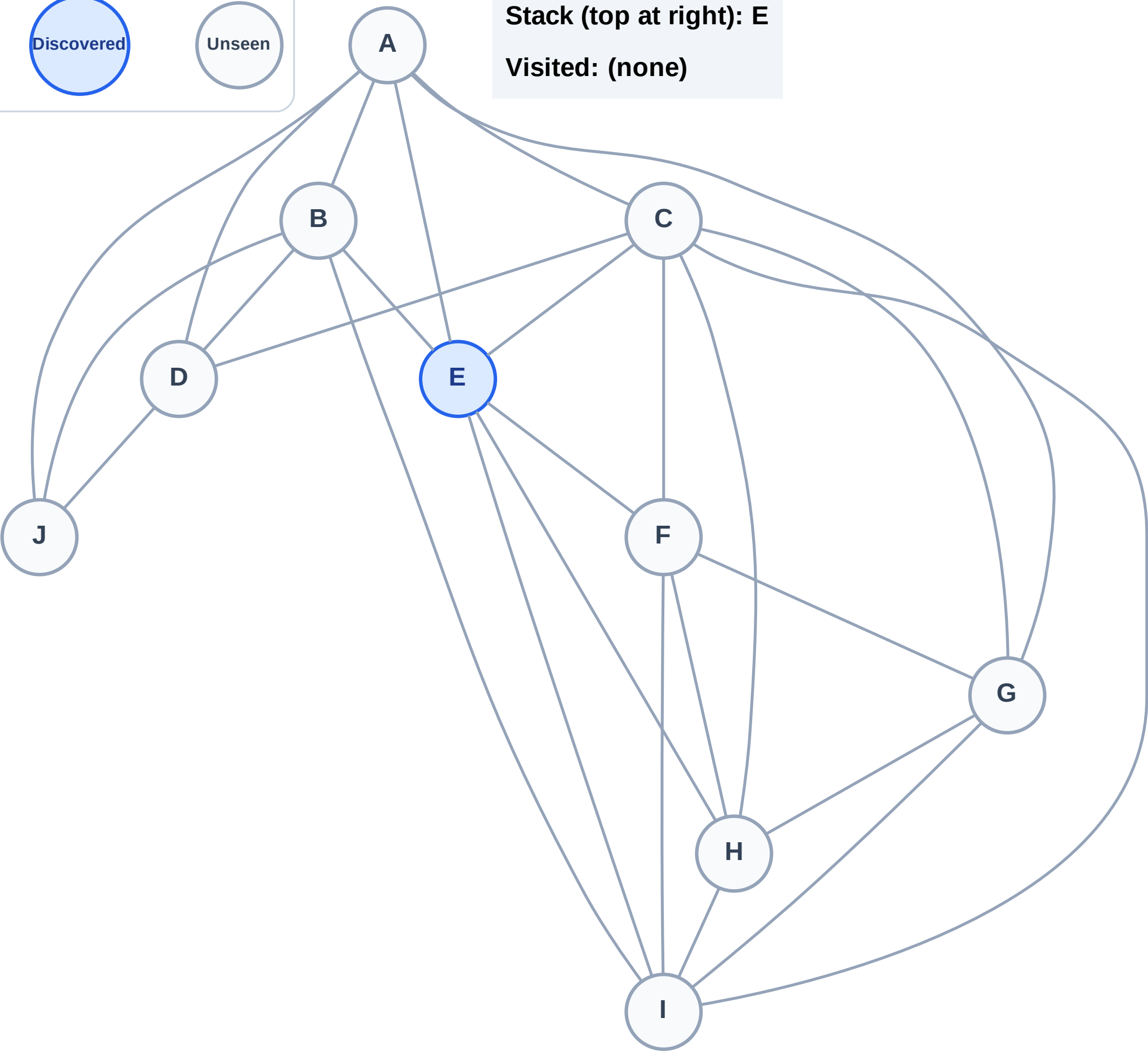
Processing

Discovered

Unseen

Stack (top at right): E

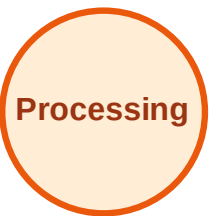
Visited: (none)



DFS Traversal

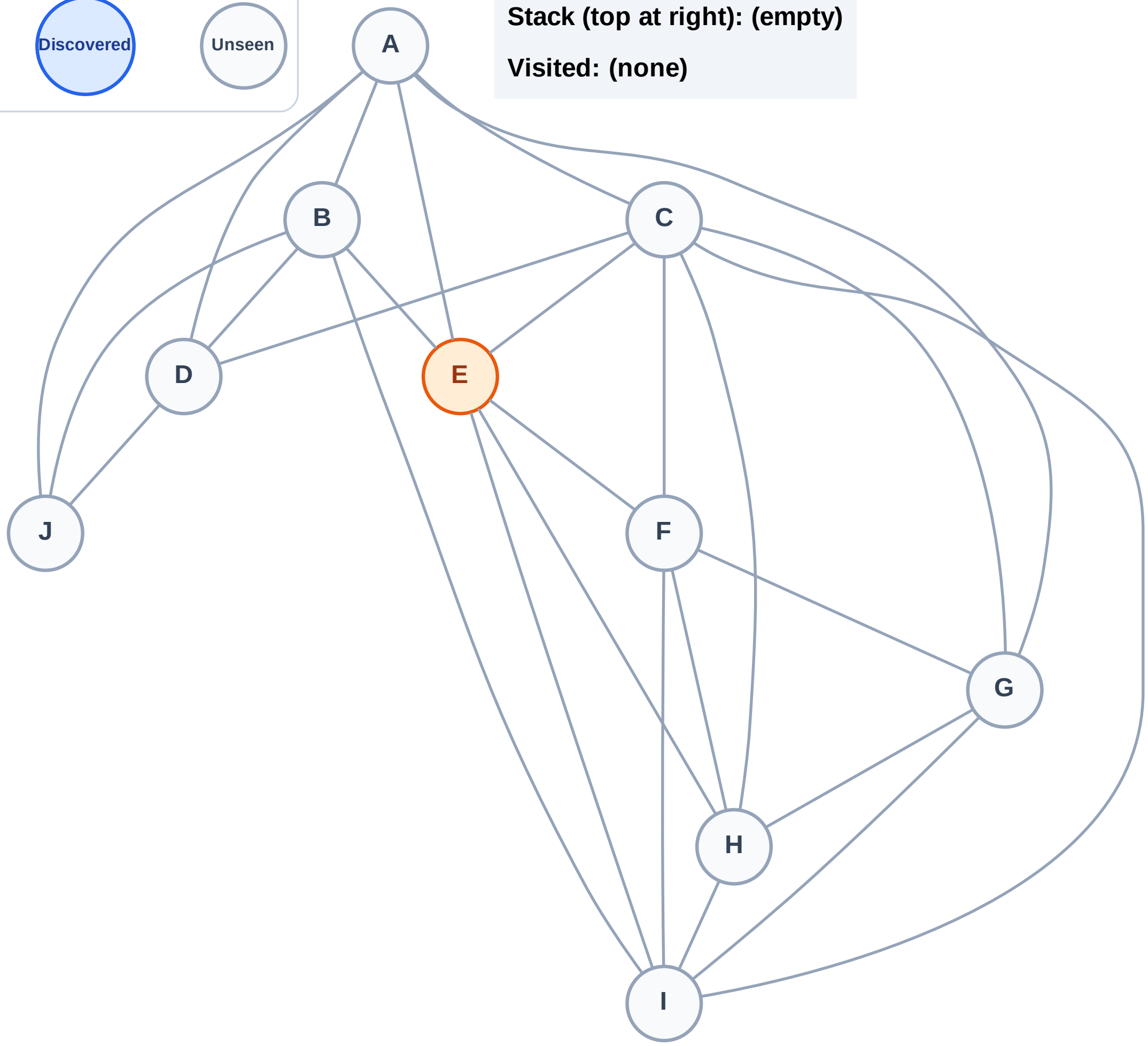
Step 2: Process node E

Legend



Stack (top at right): (empty)

Visited: (none)



DFS Traversal

Step 3: Discover I, H, F, C, B, A from E

Legend

Complete

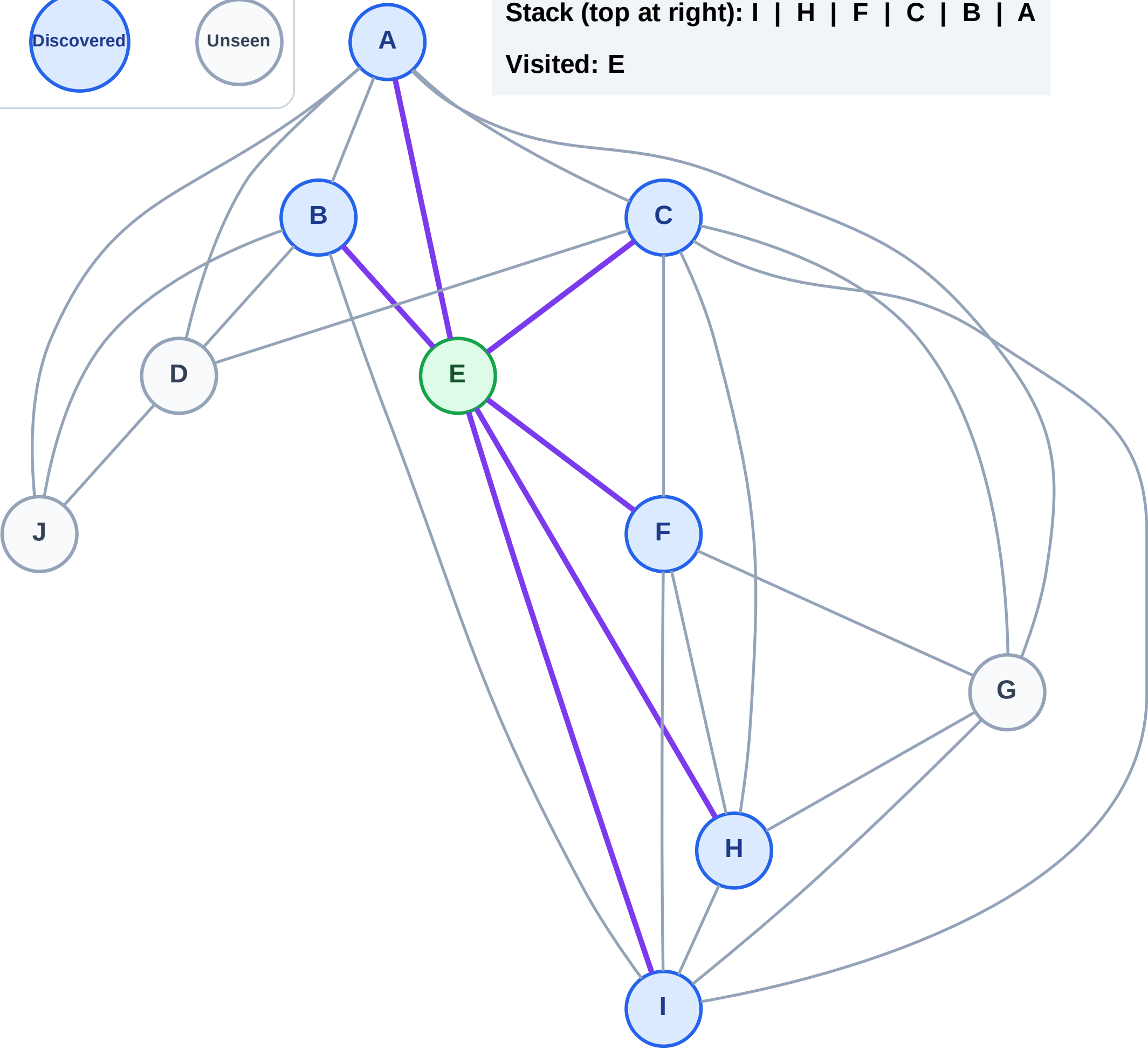
Processing

Discovered

Unseen

Stack (top at right): I | H | F | C | B | A

Visited: E



DFS Traversal

Step 4: Process node A

Legend

Complete

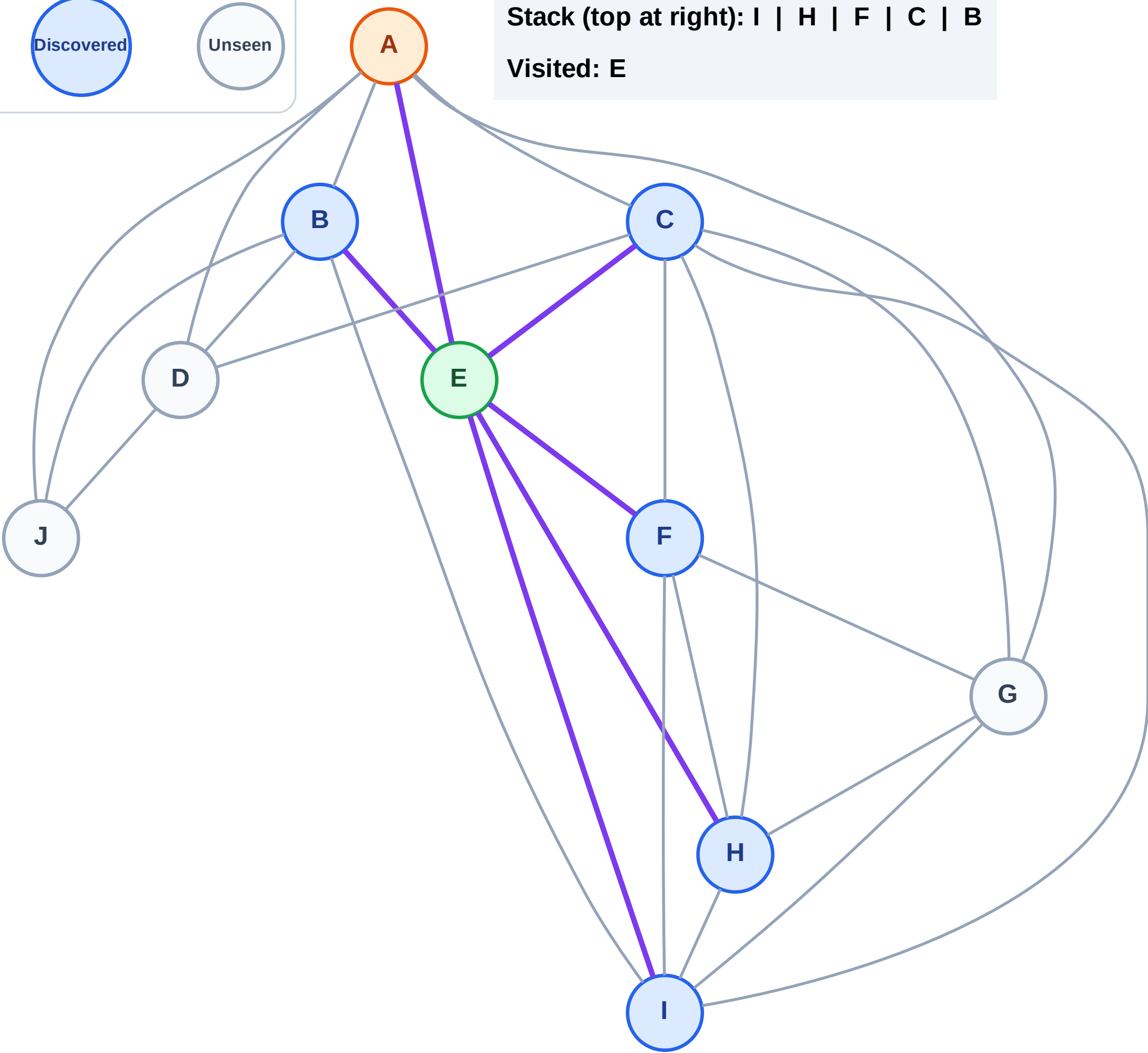
Processing

Discovered

Unseen

Stack (top at right): I | H | F | C | B

Visited: E



DFS Traversal

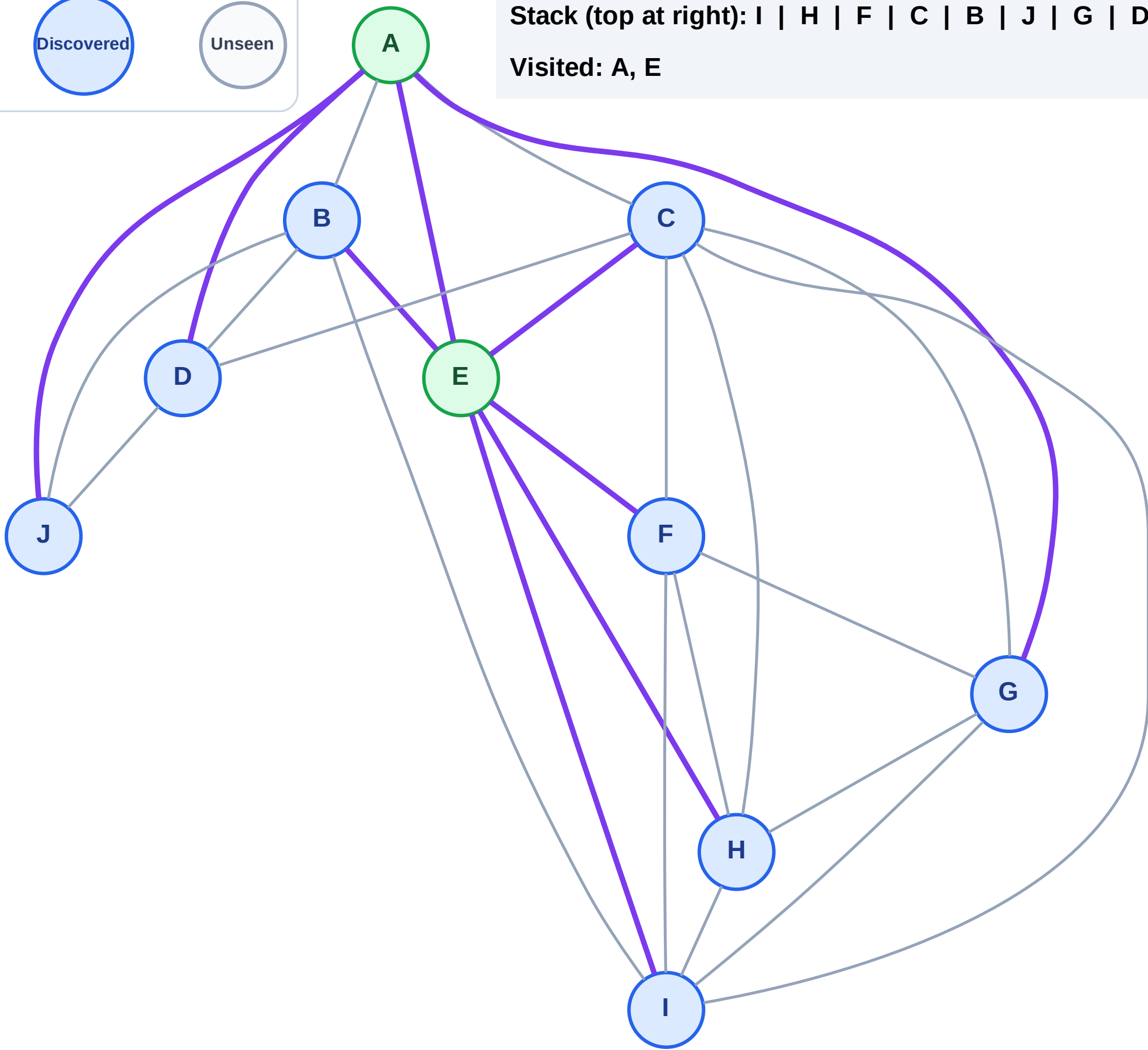
Step 5: Discover J, G, D from A

Legend



Stack (top at right): I | H | F | C | B | J | G | D

Visited: A, E



DFS Traversal

Step 6: Process node D

Legend

Complete

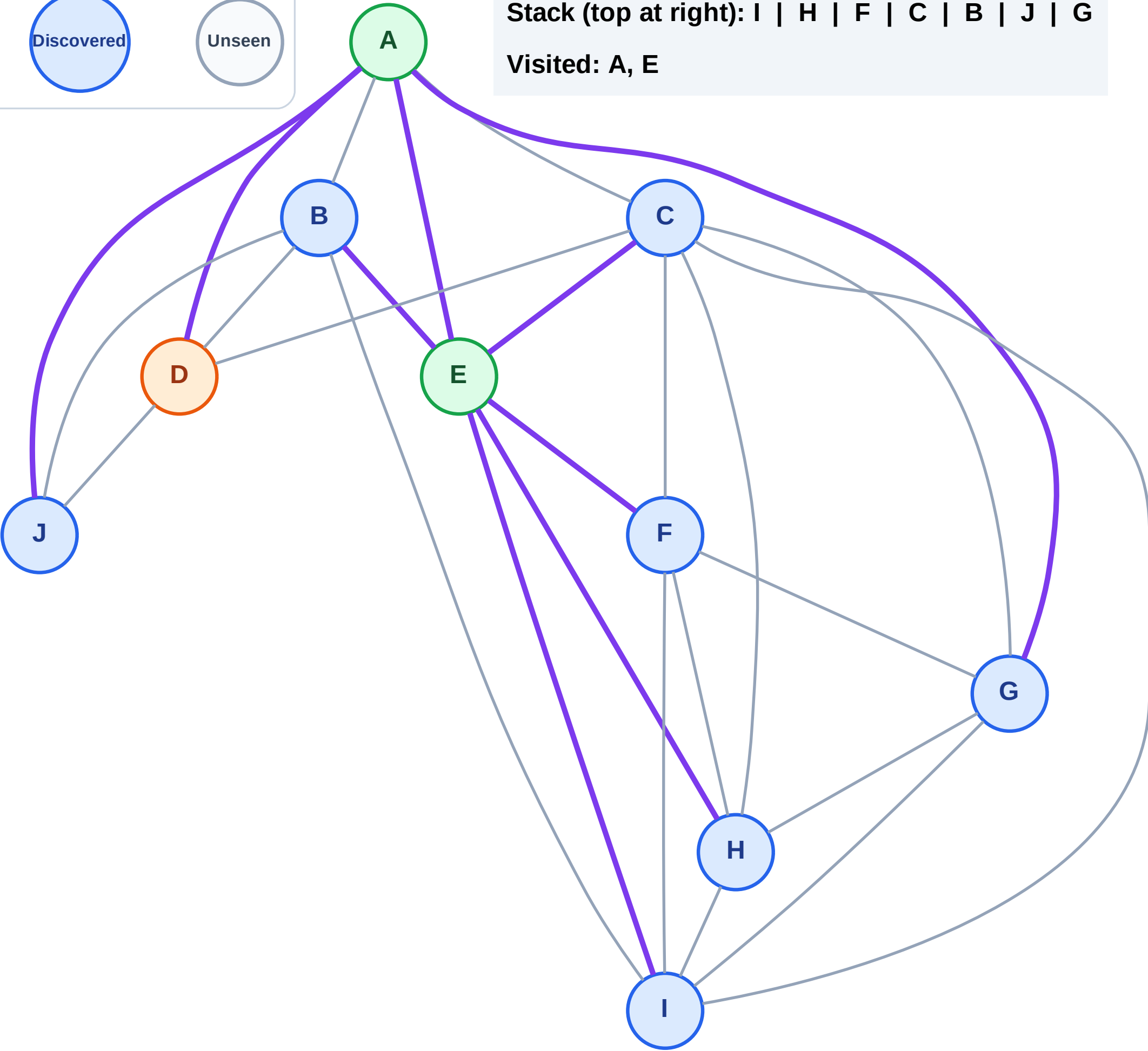
Processing

Discovered

Unseen

Stack (top at right): I | H | F | C | B | J | G

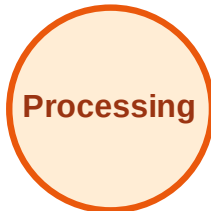
Visited: A, E



DFS Traversal

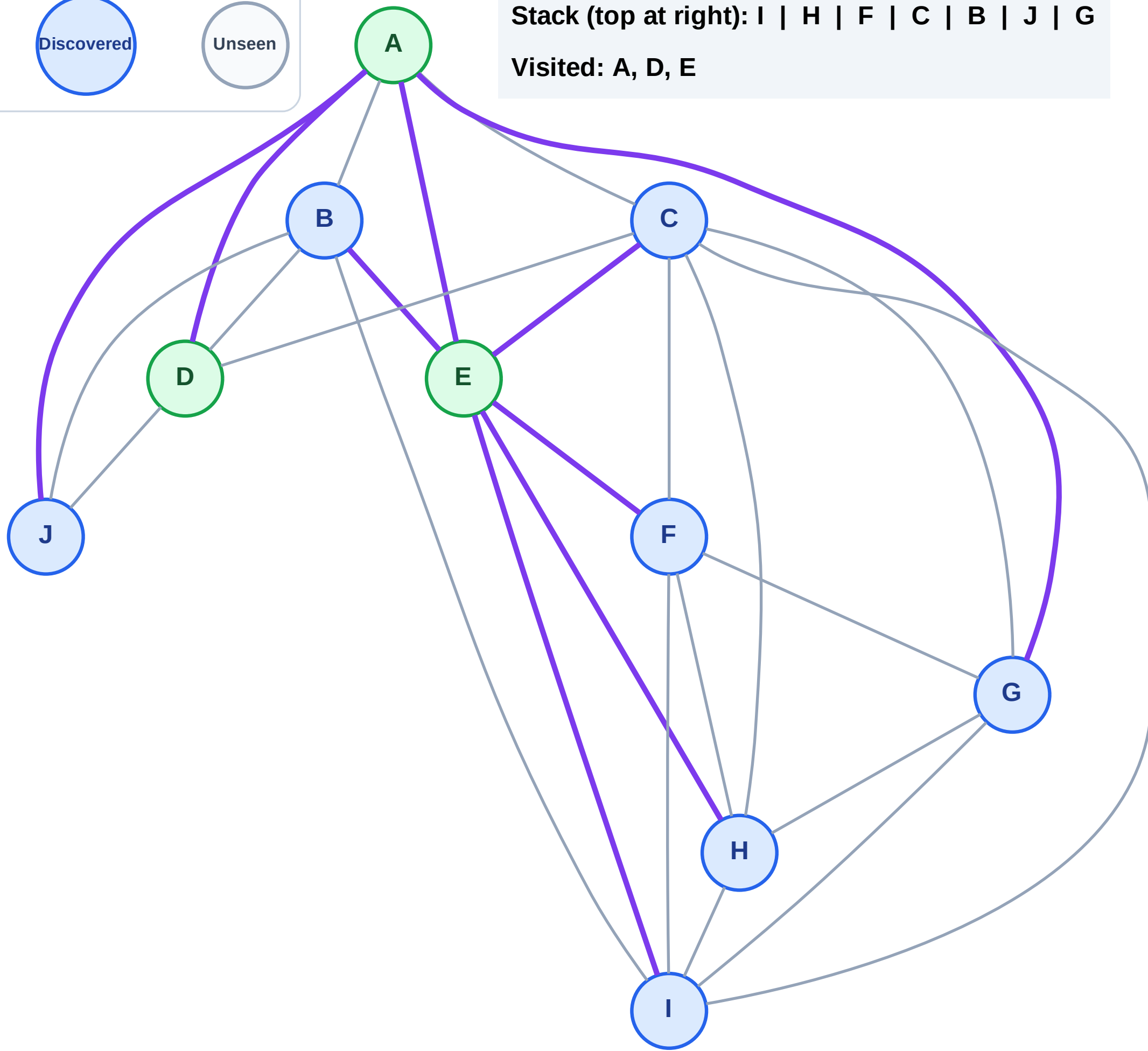
Step 7: No new neighbors from D

Legend



Stack (top at right): I | H | F | C | B | J | G

Visited: A, D, E



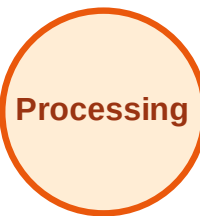
DFS Traversal

Step 8: Process node G

Legend



Complete



Processing



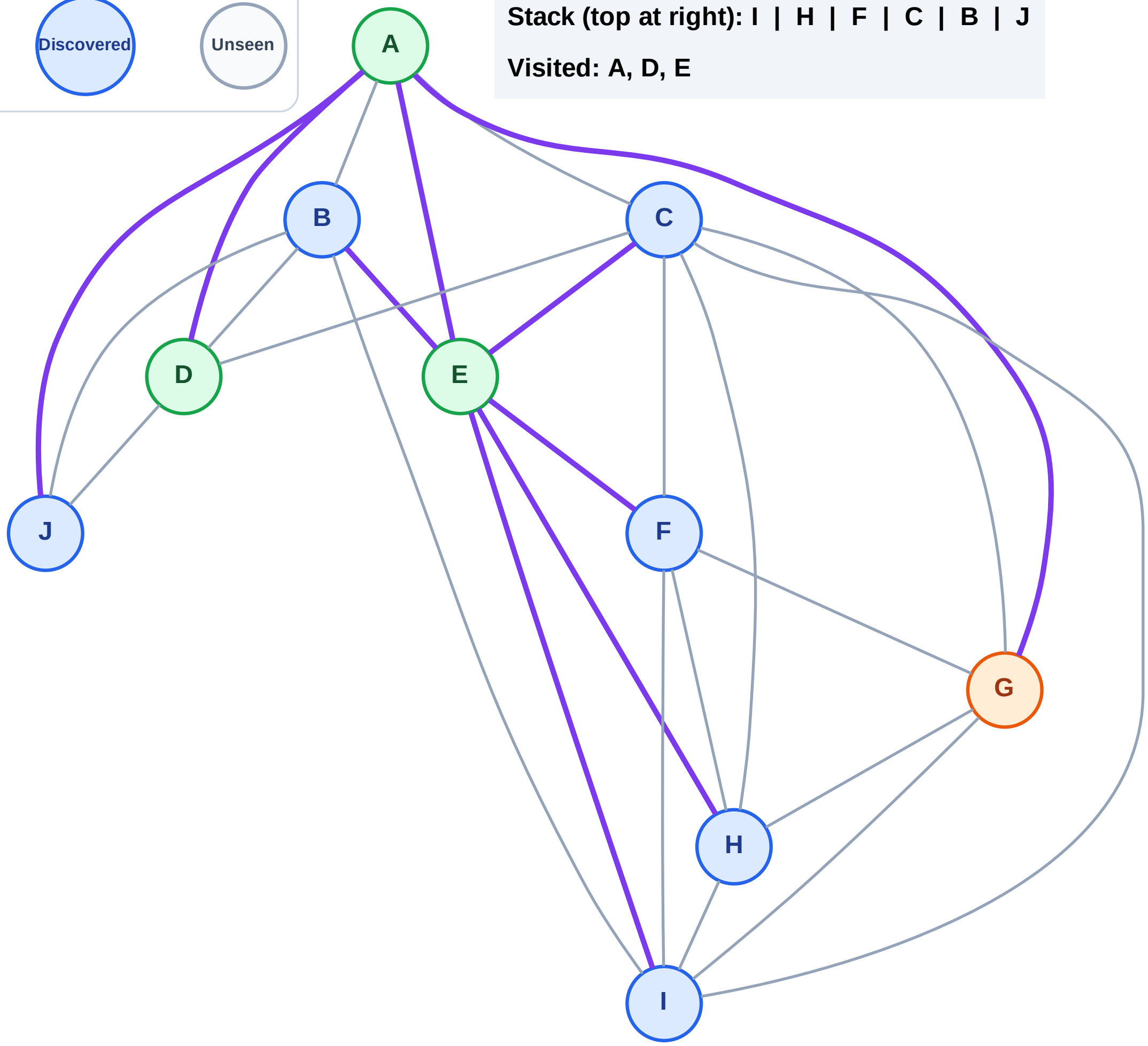
Discovered



Unseen

Stack (top at right): I | H | F | C | B | J

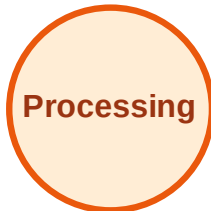
Visited: A, D, E



DFS Traversal

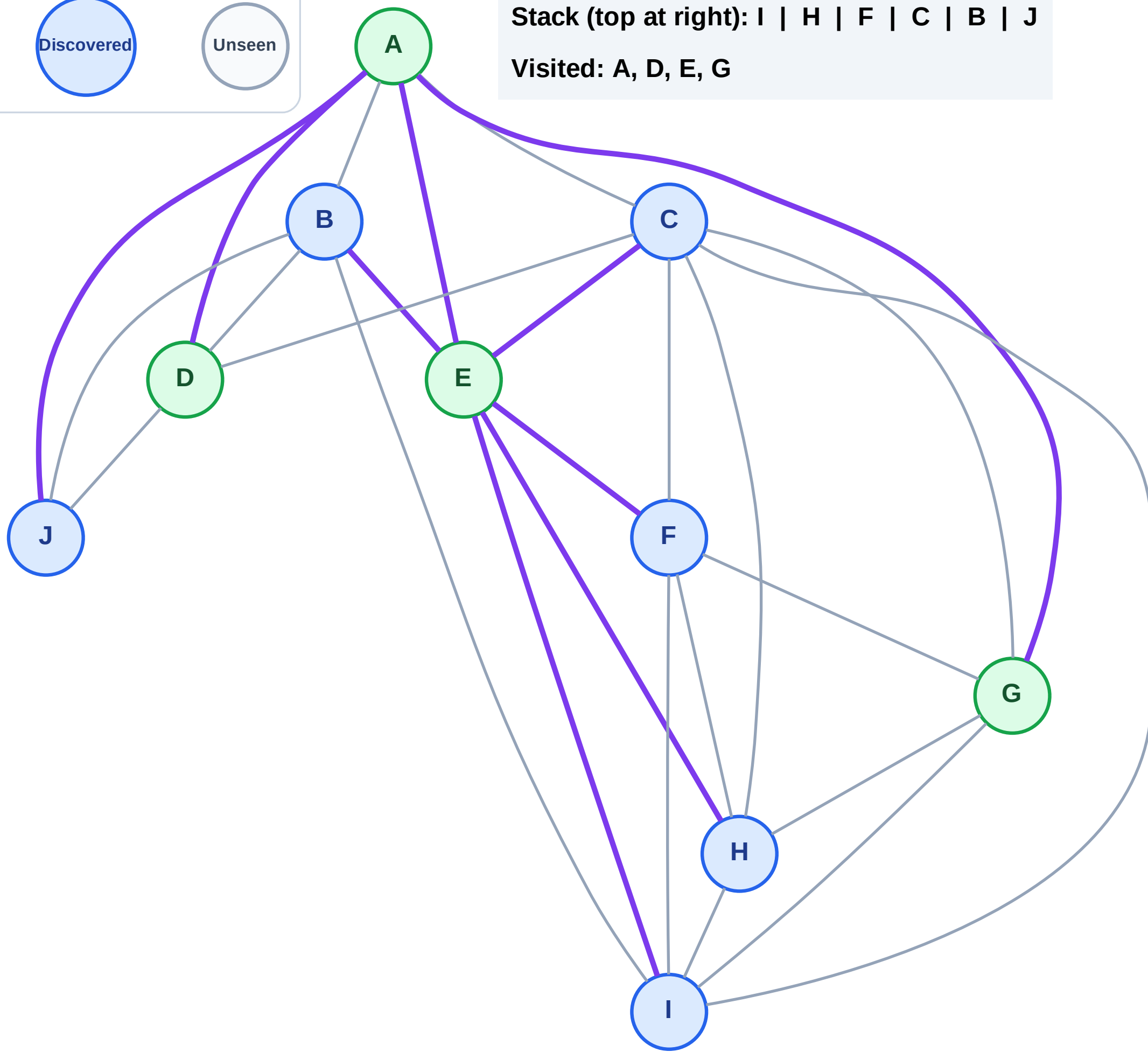
Step 9: No new neighbors from G

Legend



Stack (top at right): I | H | F | C | B | J

Visited: A, D, E, G



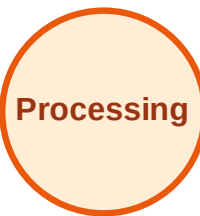
DFS Traversal

Step 10: Process node J

Legend



Complete



Processing



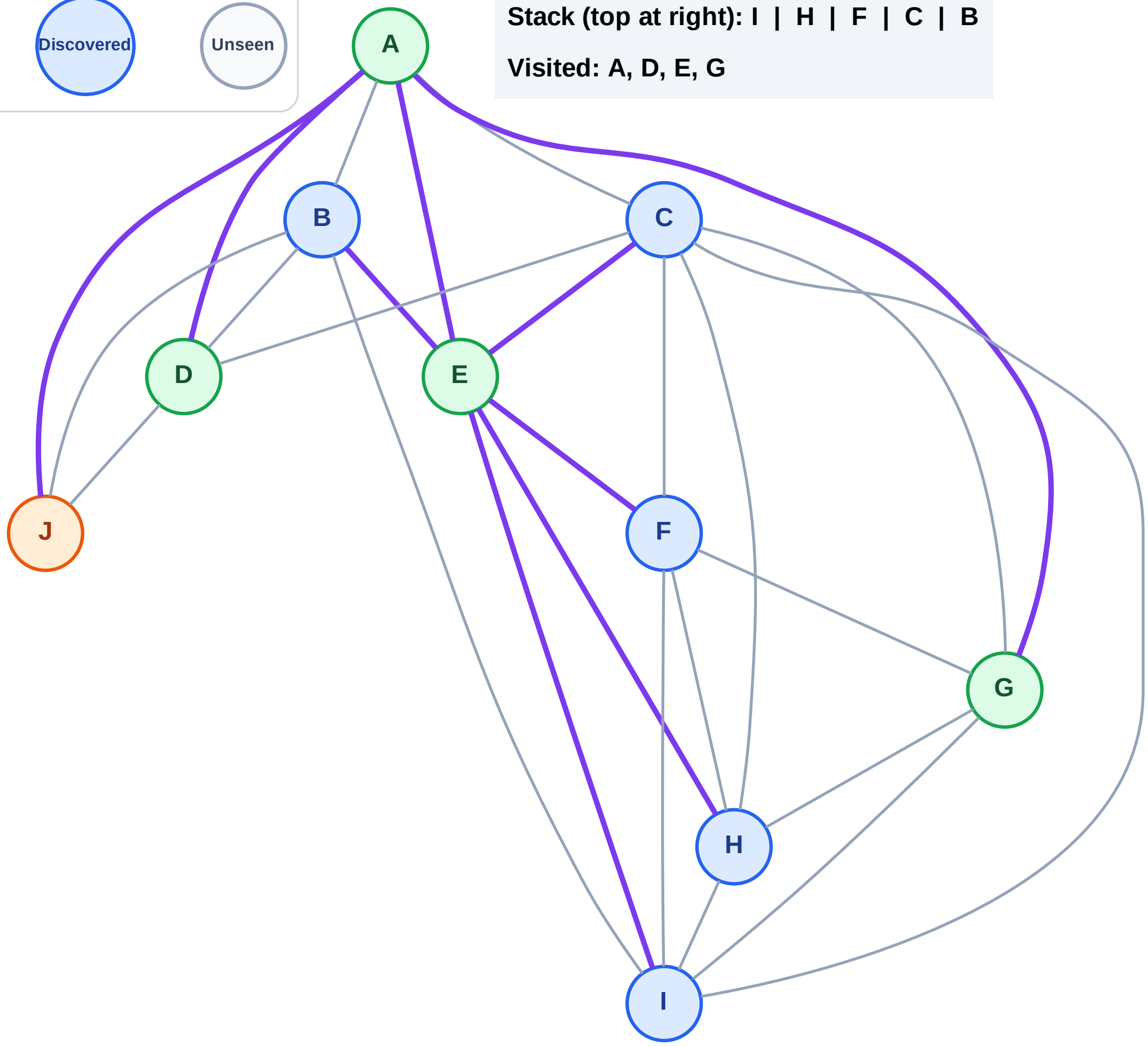
Discovered



Unseen

Stack (top at right): I | H | F | C | B

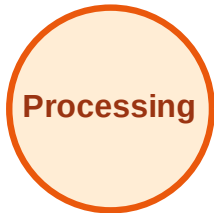
Visited: A, D, E, G



DFS Traversal

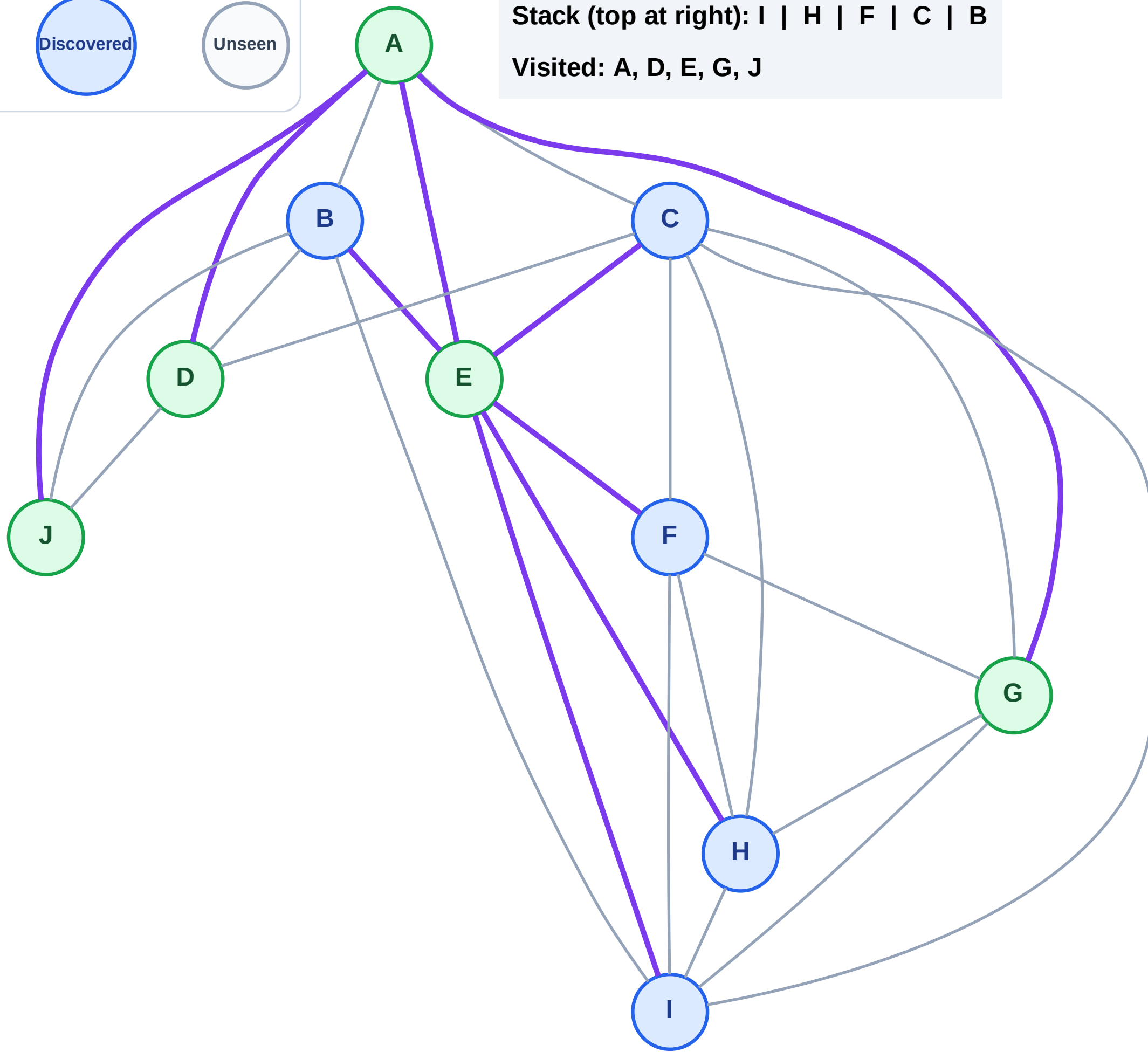
Step 11: No new neighbors from J

Legend



Stack (top at right): I | H | F | C | B

Visited: A, D, E, G, J



DFS Traversal

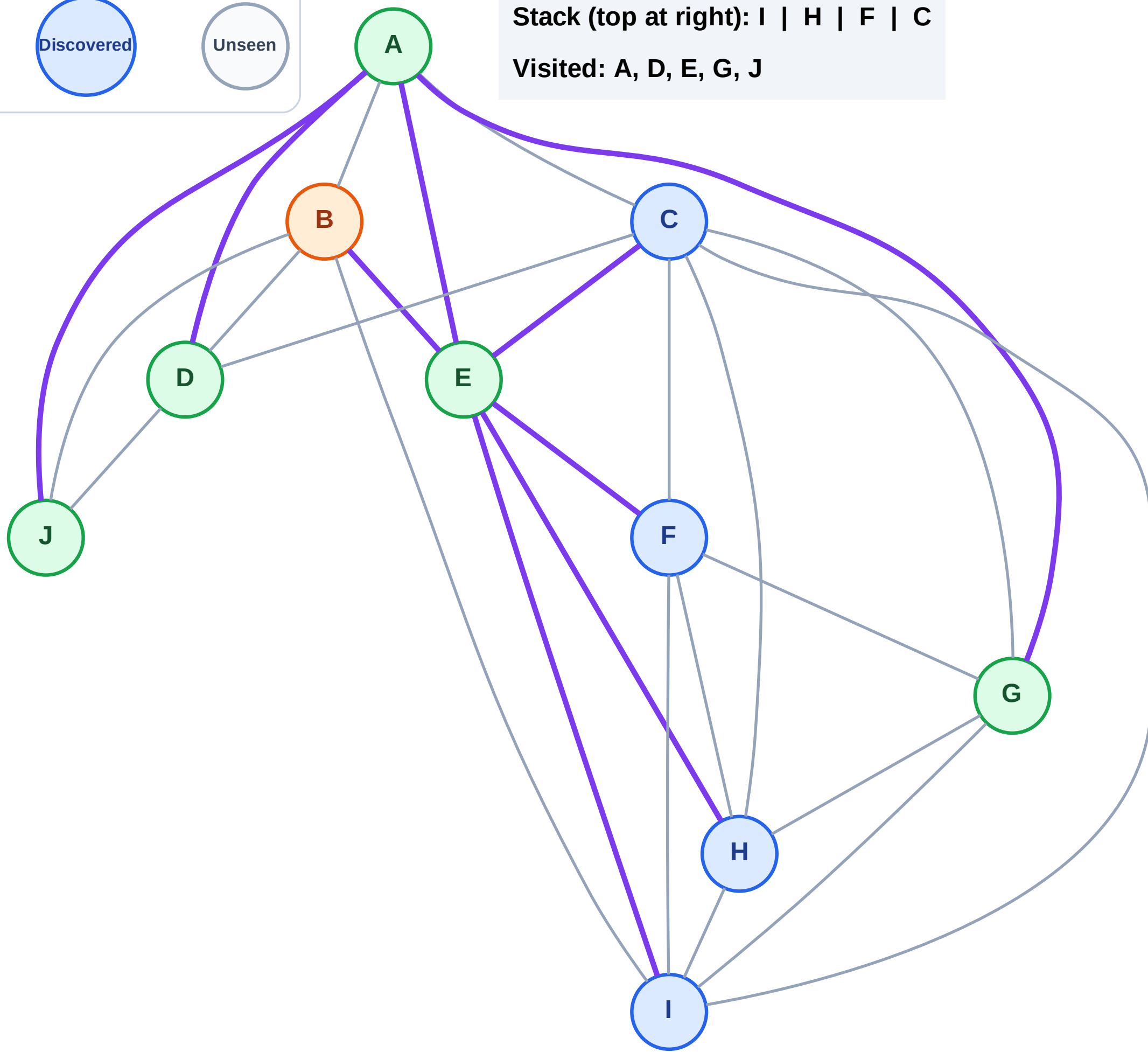
Step 12: Process node B

Legend



Stack (top at right): I | H | F | C

Visited: A, D, E, G, J



Step 13: No new neighbors from B

Stack (top at right): I | H | F | C

Legend

Processing

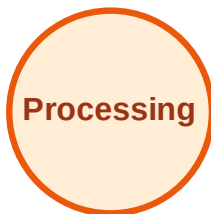
Unseen



DFS Traversal

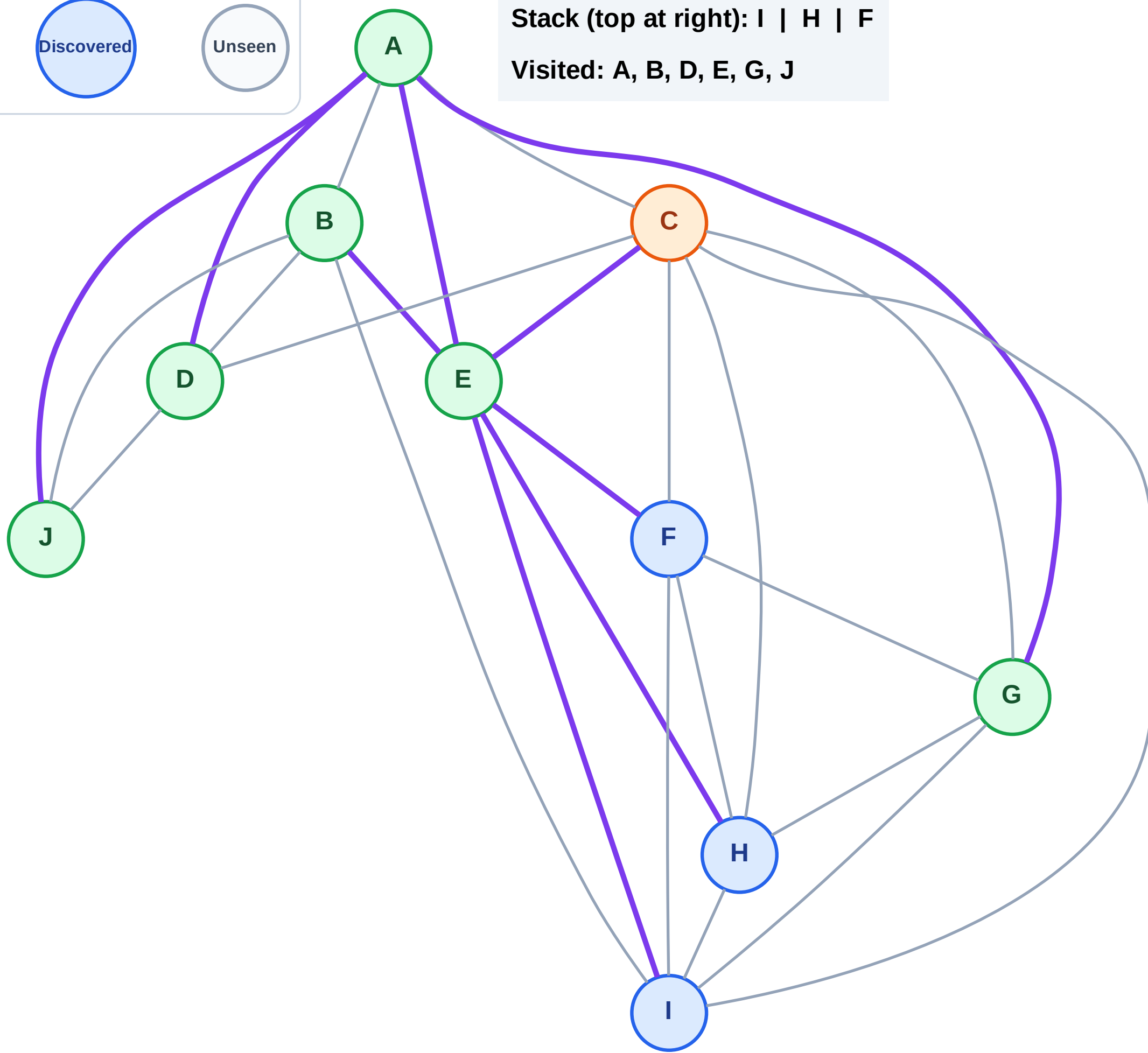
Step 14: Process node C

Legend



Stack (top at right): I | H | F

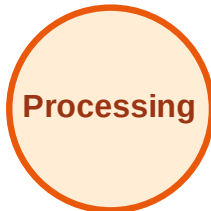
Visited: A, B, D, E, G, J



DFS Traversal

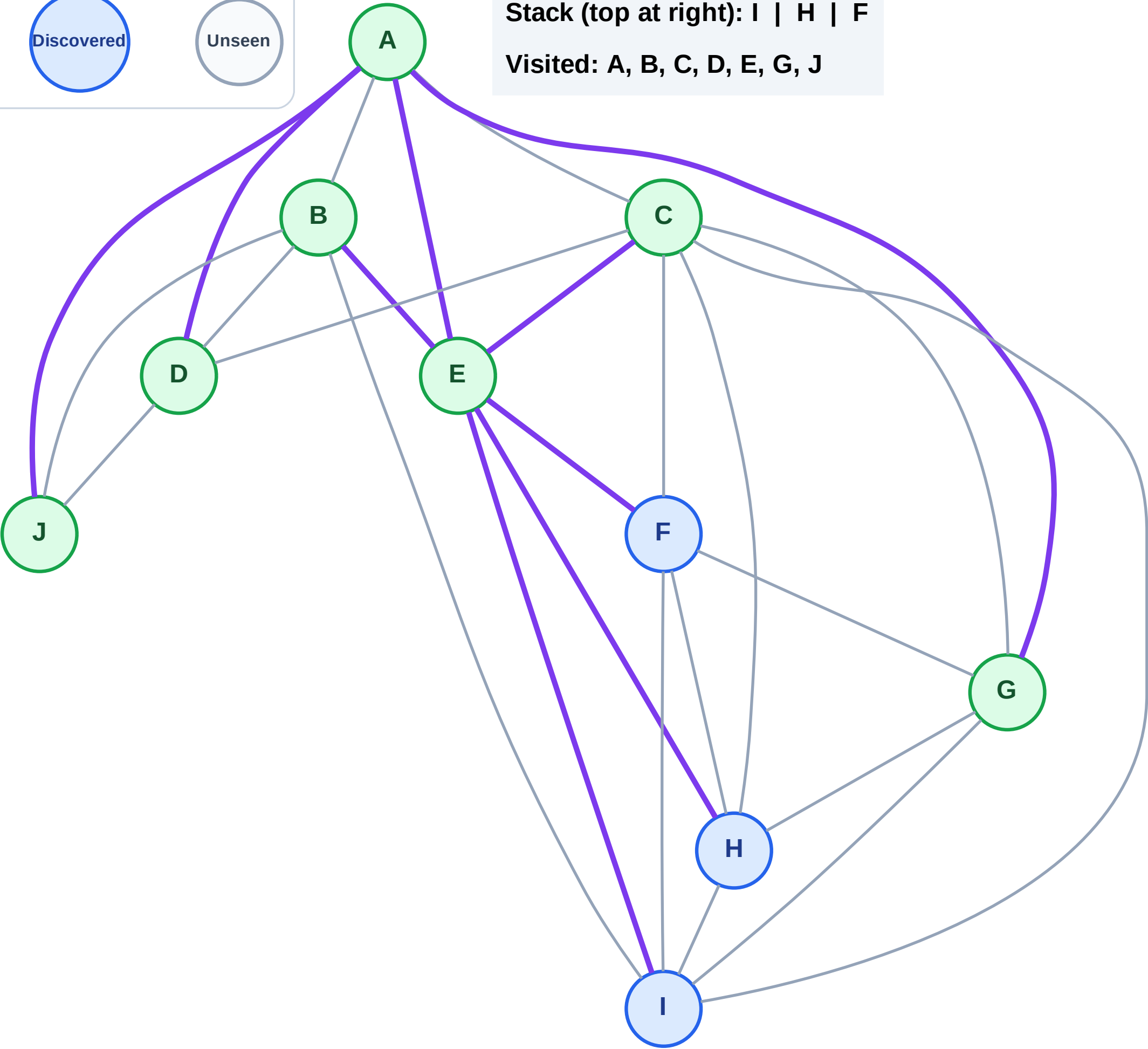
Step 15: No new neighbors from C

Legend



Stack (top at right): I | H | F

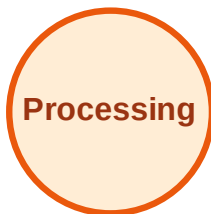
Visited: A, B, C, D, E, G, J



DFS Traversal

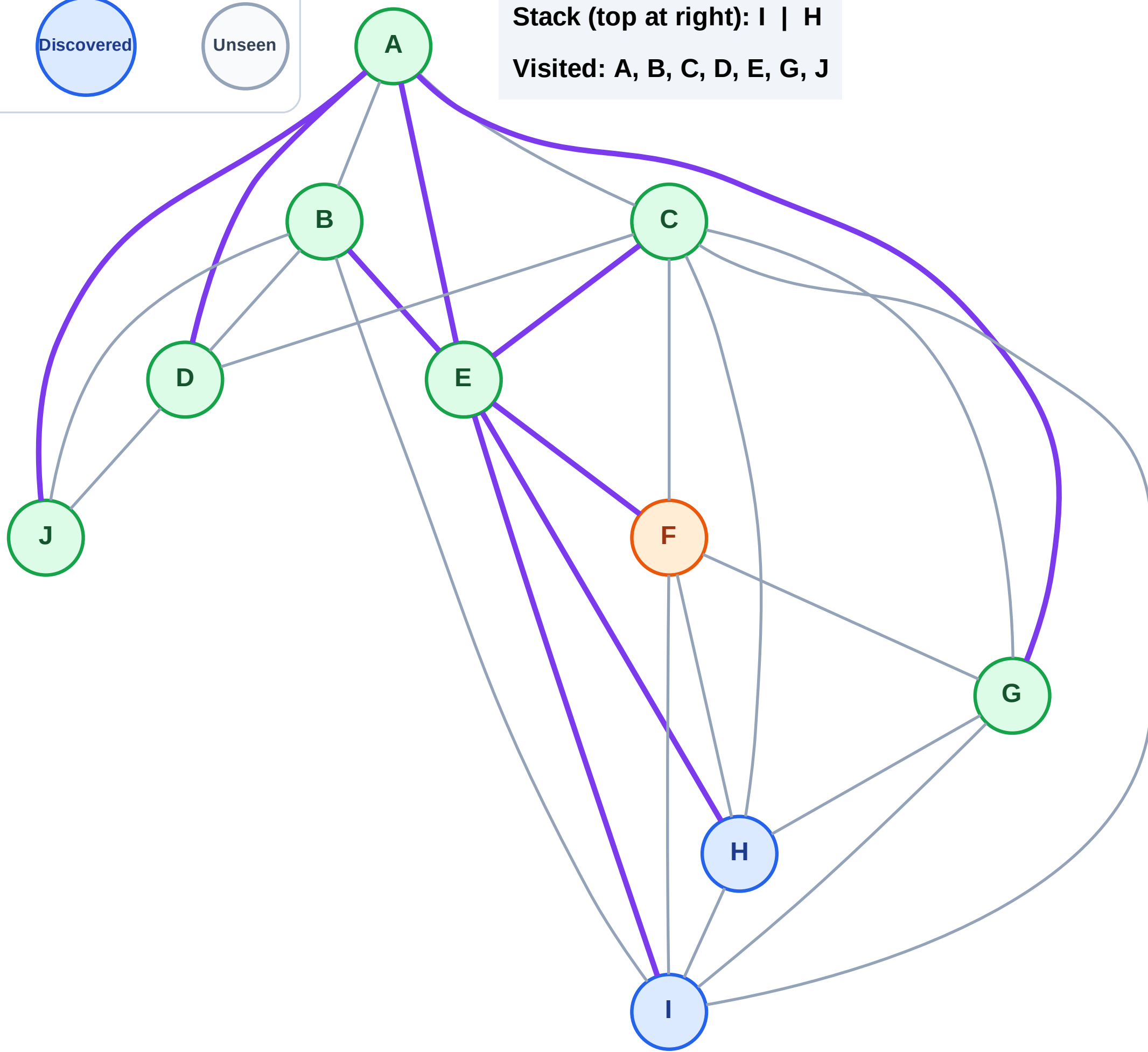
Step 16: Process node F

Legend



Stack (top at right): I | H

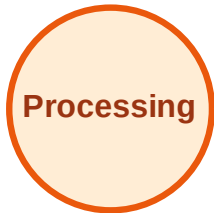
Visited: A, B, C, D, E, G, J



DFS Traversal

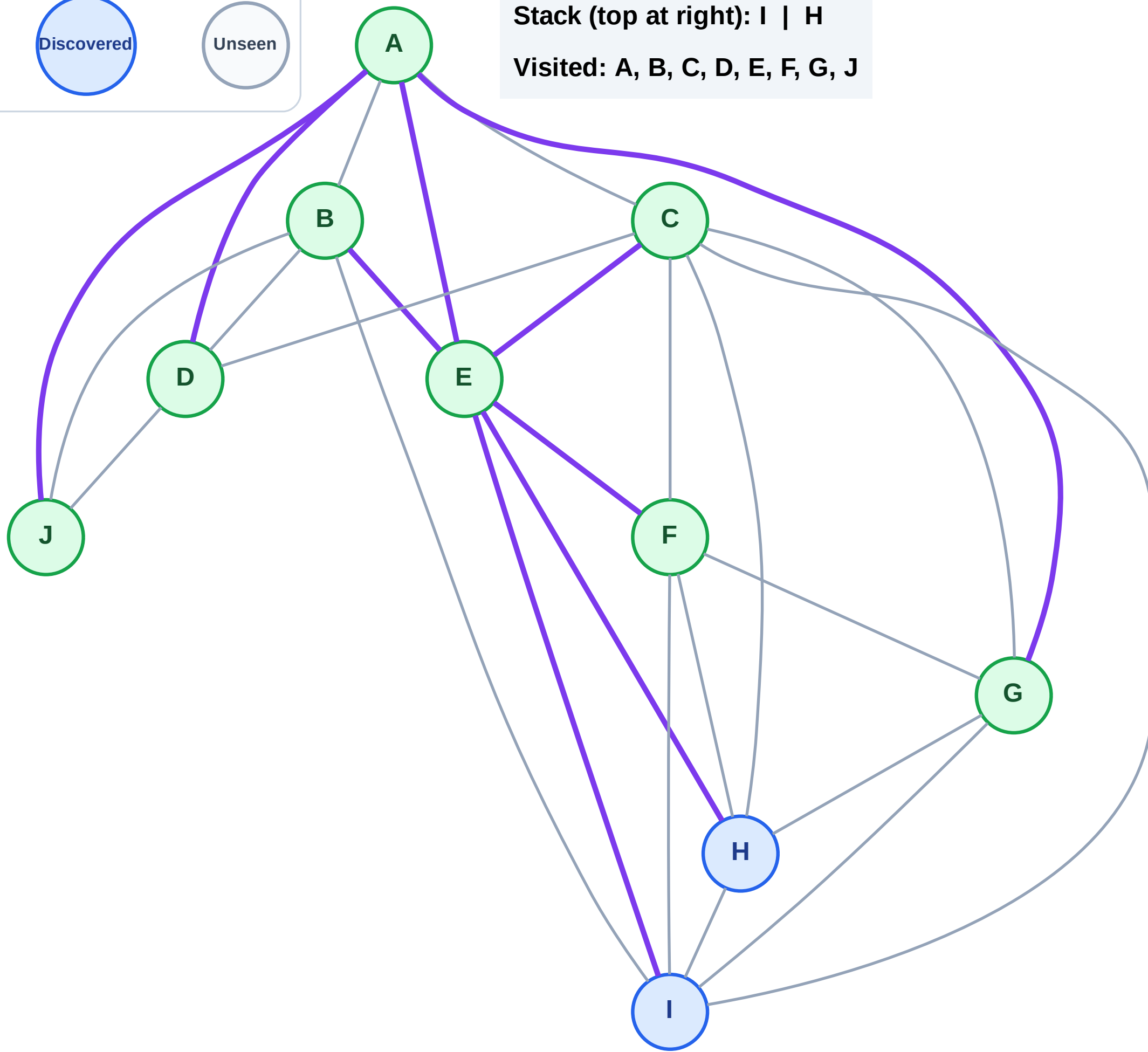
Step 17: No new neighbors from F

Legend



Stack (top at right): I | H

Visited: A, B, C, D, E, F, G, J



DFS Traversal

Step 18: Process node H

Legend

Complete

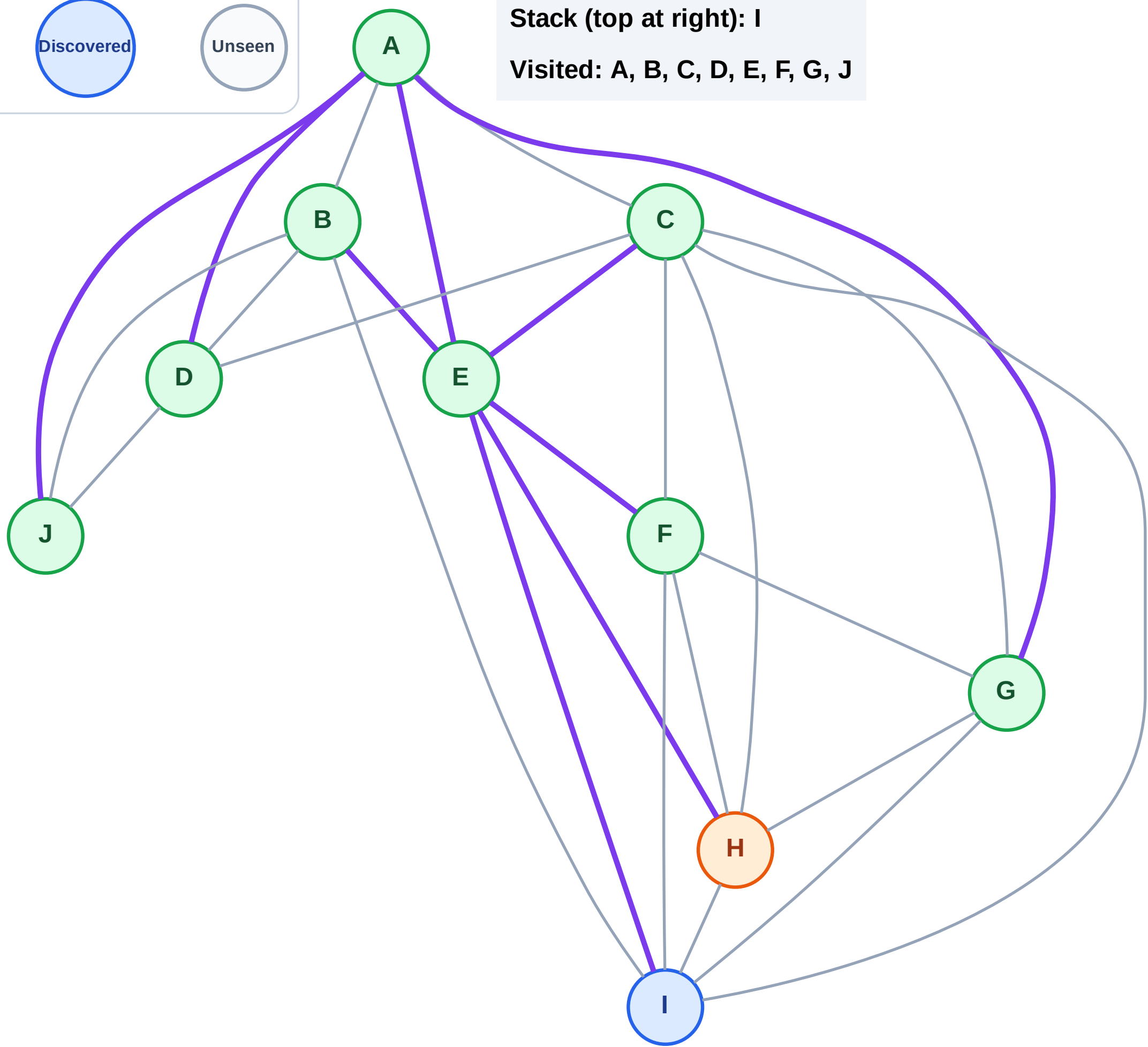
Processing

Discovered

Unseen

Stack (top at right): I

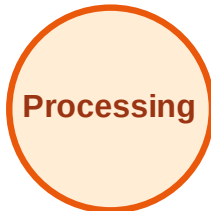
Visited: A, B, C, D, E, F, G, J



DFS Traversal

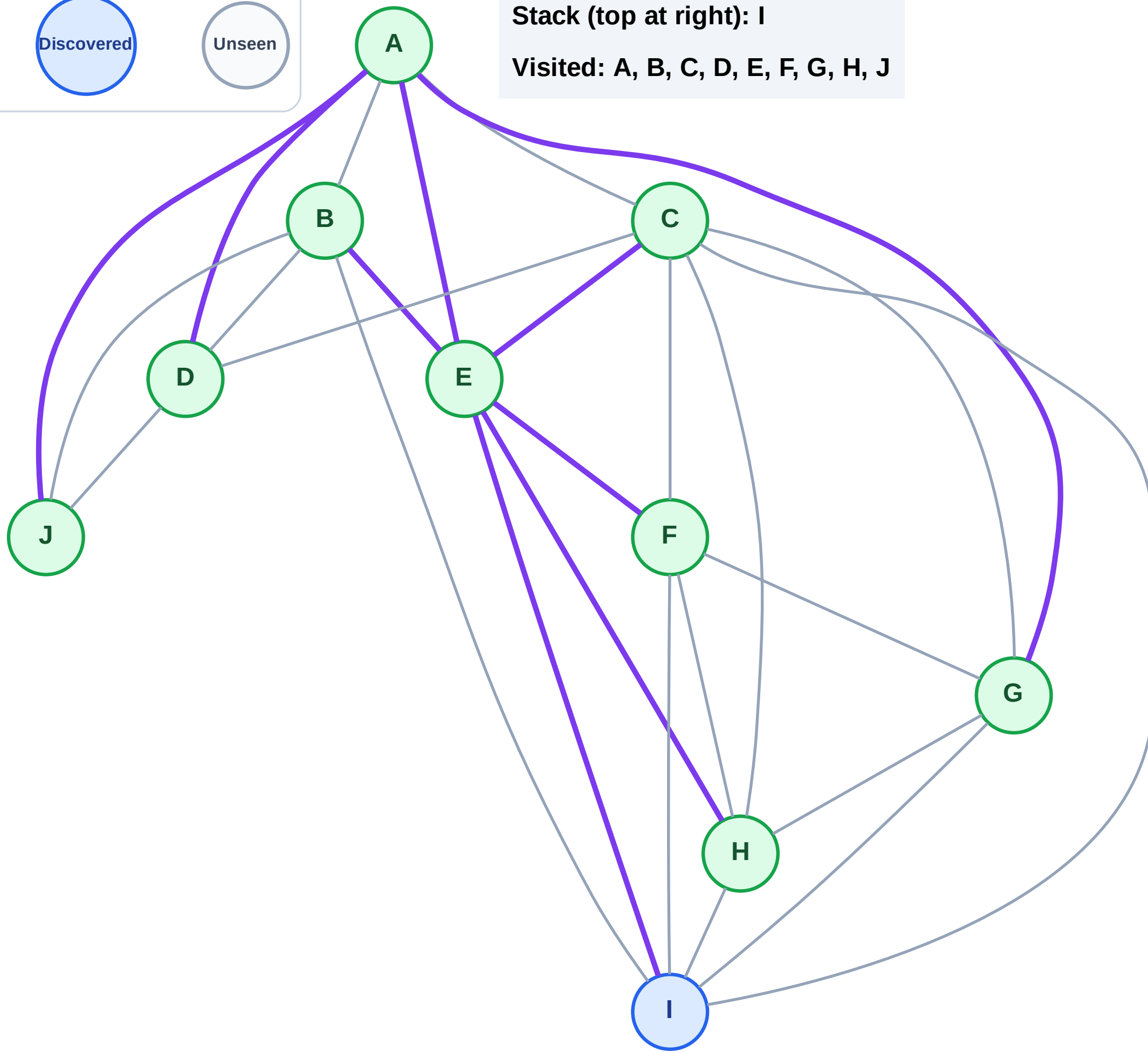
Step 19: No new neighbors from H

Legend



Stack (top at right): I

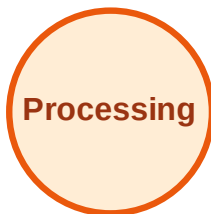
Visited: A, B, C, D, E, F, G, H, J



DFS Traversal

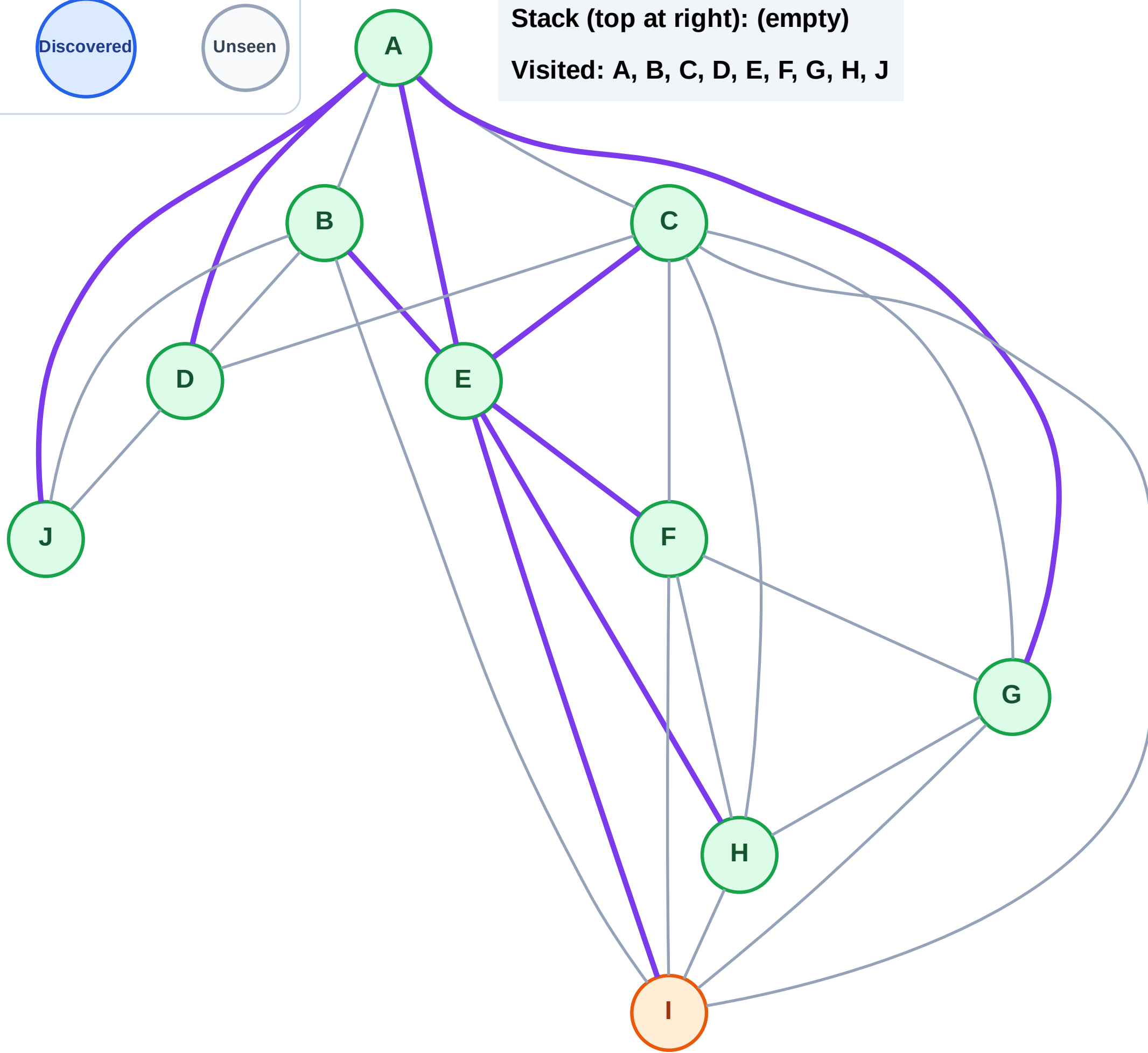
Step 20: Process node I

Legend



Stack (top at right): (empty)

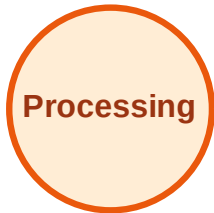
Visited: A, B, C, D, E, F, G, H, J



DFS Traversal

Step 21: No new neighbors from I

Legend



Stack (top at right): (empty)

Visited: A, B, C, D, E, F, G, H, I, J

